



EUROPEAN
COMMISSION

Brussels, 8.10.2026
C(2026) 6899 final

ANNEX

ANNEX

to the

Commission Implementing Decision

**laying down rules for the application of Directive 2010/75/EU of the European
Parliament and of the Council as regards uniform conditions for operating rules for
rearing of poultry and pigs**

{SWD(2026) 298 final}

ANNEX

UNIFORM CONDITIONS FOR OPERATING RULES FOR LIVESTOCK (PIGS AND POULTRY REARING)

SCOPE

These Uniform Conditions (UC) concern the activities specified in Annex Ia to Directive 2010/75/EU (the IED).

The Livestock Unit (LSU) level of an installation is to be calculated using the conversion rates given in Annex Ia to the IED.

In particular, the UC cover the following on-farm processes and activities:

- (a) nutritional management;
- (b) feed preparation;
- (c) housing;
- (d) manure management, including collection, storage, processing and landspreading;
- (e) storage of dead animals.

The UC apply in parallel to other relevant legislation, e.g. Council Directive 91/676/EEC¹, Directive (EU) 2016/2284 of the European Parliament and of the Council², Regulation (EC) No 1069/2009 of the European Parliament and of the Council³, legislation on animal welfare (e.g. Council Directive 2008/120/EC⁴, Council Directive 98/58/EC⁵, Council Directive 1999/74/EC⁶, Council Directive 2007/43/EC⁷).

¹ Council Directive 91/676/EEC (1) of 12 December 1991 concerning the protection of waters against pollution caused by nitrates from agricultural sources (OJ L 375/1, 31.12.1991, p.1, ELI: <http://data.europa.eu/eli/dir/1991/676/oj>).

² Directive (EU) 2016/2284 of the European Parliament and of the Council of 14 December 2016 on the reduction of national emissions of certain atmospheric pollutants, amending Directive 2003/35/EC and repealing Directive 2001/81/EC (OJ L 344, 17.12.2016, pp. 1, ELI: <http://data.europa.eu/eli/dir/2016/2284/oj>).

³ Regulation (EC) No 1069/2009 of the European Parliament and of the Council of 21 October 2009 laying down health rules as regards animal by-products and derived products not intended for human consumption and repealing Regulation (EC) No 1774/2002 (Animal by-products Regulation) (OJ L 300/1, 14.11.2009, p.1, ELI: <http://data.europa.eu/eli/reg/2009/1069/oj>).

⁴ Council Directive 2008/120/EC of 18 December 2008 laying down minimum standards for the protection of pigs (OJ L 47, 18.2.2009, p. 5–13, ELI: <http://data.europa.eu/eli/dir/2008/120/oj>).

⁵ Council Directive 98/58/EC of 20 July 1998 concerning the protection of animals kept for farming purposes (OJ L 221, 8.8.1998, p. 23-27, ELI: <http://data.europa.eu/eli/dir/1998/58/oj>).

⁶ Council Directive 1999/74/EC of 19 July 1999 laying down minimum standards for the protection of laying hens; (OJ L 203, 3.8.1999, pp. 53–57, ELI: <http://data.europa.eu/eli/dir/1999/74/oj>).

⁷ Council Directive 2007/43/EC of 28 June 2007 laying down minimum rules for the protection of chickens kept for meat production (OJ L 182, 12.7.2007, pp. 19–28, ELI: <http://data.europa.eu/eli/dir/2007/43/oj>).

DEFINITIONS

For the purposes of this Annex, the following definitions apply:

General terms	
Term used	Definition
Ad libitum	The provision of free access to feed or water thereby allowing the animal to self-regulate feed or water intake according to its biological needs.
Animal house	A plant on a farm specifically designed to house livestock.
Animal place	Space provided per animal in a housing system taking into account the maximum capacity of the plant.
Conservation tillage	Any method of soil cultivation that leaves the previous year's crop residue (such as corn stalks or wheat stubble) on the surface of fields before and after planting the next crop, to reduce soil erosion and run-off.
Earth-banked slurry storage	Storage of slurry in in-ground structures (e.g. lagoons, basins) constructed within compacted earth banks.
Existing farm	A farm which is not a new farm.
Existing plant	A plant which is not a new plant.
Farm	An installation as defined in Article 3 point (3) of the IED, where pigs or poultry are reared.
Loose housing systems for farrowing sows	Housing systems for farrowing sows providing larger areas for movement compared to fixed crate pens, to ensure enhanced animal welfare standards, including sufficient space to allow the sow to turn around and express natural behaviours with minimum space levels per sow.
Manure	Slurry and/or solid manure.
Newly covered farm	An existing farm carrying out activities referred to in Annex Ia and which did not fall in the scope of the IED before 4 August 2024.
New farm	A farm whose construction has been authorised or registered after the applicable deadline set out in Article 3(5) of Directive (EU) 2024/1785 or a complete replacement of a farm after the applicable deadline set out in Article 3(5) of that Directive.
New plant	A plant whose construction at the site of the farm has been authorised or registered after the applicable deadline set out in Article 3(5) of Directive (EU) 2024/1785 or a complete replacement of a plant on the existing foundations, after the applicable deadline set out in Article 3(5) of that Directive.
On-farm	Within the premises of the IED installation.
Organic production	Organic production defined in accordance with Regulation (EU) 2018/848
Plant	A part of the farm where one of the following processes or activities is carried out: animal housing, manure storage, manure processing. A plant consists of a single building (or facility) and/or the necessary equipment to carry out processes or activities.
Sensitive receptor	Areas which need special protection, such as: (a) residential areas; (b) areas where human activities are carried out (e.g. neighbouring workplaces, schools, day-care centres, recreational areas, hospitals or nursing homes); (c) sensitive ecosystems/habitats.
Slurry	Faeces and urine mixed or not with some litter material and some water to give a liquid manure that flows under gravity and can be pumped.
Slurry stores	Fixed structures designed to hold slurry.
Solid manure	Faeces or droppings and urine mixed or not with litter material that do not flow under gravity and cannot be pumped.

Waste water	Rainwater run-off commonly mixed with manure, water derived from the cleaning of surfaces (e.g. floors) and equipment, and water derived from the operation of air cleaning systems. For the purposes of this document, effluents from other uses (e.g. black water / sewage, waste water from cleaning of on-farm storage of dead animals, water contaminated with fuels / oil) are not considered.
-------------	--

Pollutants and parameters	
Term used	Definition
Total ammoniacal nitrogen	Ammonium-N (NH ₄ -N) and its compounds, including uric acid, which are readily broken down to NH ₄ -N.
Total nitrogen	Total nitrogen, expressed as N, includes free ammonia and ammonium (NH ₄ -N), nitrites (NO ₂ -N), nitrates (NO ₃ -N) and organic nitrogen compounds.
Total nitrogen excreted	Total nitrogen eliminated from animal metabolic processes through urine and faeces.
Total phosphorus	Total phosphorus, expressed as P ₂ O ₅ , includes all inorganic and organic phosphorus compounds, dissolved or bound to particles.
Total phosphorus excreted	Total phosphorus eliminated from animal metabolic processes through urine and faeces.
Animal categories	
Term used	Definition
Breeders	Parent poultry stock (males and females) kept for reproductive purposes.
Broilers	Chickens reared for meat production, not including breeders.
Ducks	Birds of the <i>Anas</i> and <i>Cairina</i> genus, reared for meat or egg production, not including breeders.
Farrowing sows	Female pig between the perinatal period and the weaning of the piglets.
Fattening pigs	Production pigs typically reared from a live weight of 30 kg to slaughter or first service. This category includes growers, finishers and gilts that have not been serviced.
Gestating sows	Pregnant sows, including gilts.
Gilts	Female pig after puberty and before farrowing.
Mating sows	Sows ready for service and before gestation.
Heavy pigs	Pigs with a live slaughter weight typically above 130 kg, due to extended production cycles.
Pig	An animal of the porcine species of any age, kept for breeding or fattening.
Piglets	Pigs from birth to weaning (typically until a live weight of around 8 kg).
Poultry	Fowl (chickens), turkeys, guinea fowl, ducks, geese, quails, pigeons, pheasants and partridges reared or kept in captivity for breeding, the production of meat or eggs for consumption, or for restocking supplies of game.
Laying hens	Grown female chickens when they start laying eggs after 16 to 20 weeks of age, not including breeders, nor pullets.
Pullets	When chickens are reared for breeding, pullets are young female and male chickens until 20 weeks of age. When chickens are reared for egg production, pullets are hens before they begin to lay eggs at 16 to 20 weeks of age.
Sows	Female pigs during the rearing periods of mating, gestating and farrowing.
Turkeys	Birds of the <i>Meleagris</i> genus reared for meat production, not including breeders.
Weaners	Young pigs reared from weaning until fattening, typically reared from a live weight of around 8 kg until a live weight of around 30 kg.

GENERAL CONSIDERATIONS

These Uniform Conditions for operating rules are consistent with the use of best available techniques for the activities listed in Annex Ia to the IED. This document also includes information on emerging techniques, in accordance with Article 70i of the IED.

The best available techniques and emerging techniques in this document are neither prescriptive nor exhaustive. Other techniques may be used that ensure at least an equivalent level of environmental protection, including human health and climate protection.

Information on the environmental performance of the techniques listed in Section 4 and Section 5 (e.g. efficiency for ammonia emissions reduction) is based on the information exchange pursuant to Article 70i of the IED, and on recognised international guidance (e.g. UNECE guidance document on 'Options for ammonia mitigation'). Additional information on the environmental performance of the techniques or their combinations may be taken from Union, national or international sources (e.g. UNECE guidance documents or EMEP/EEA air pollutant emission inventory guidebook) that ensure the provision of data of an equivalent scientific quality.

It is for Member States to validate, based on scientifically established methods:

- (a) the use of techniques not listed in this document that ensure at least an equivalent level of environmental protection, including human health and climate protection, also taking into account animal welfare systems recognised at Union, national or international level;
- (b) the use of environmental performance of techniques listed in this document that are different to those in Sections 4 and 5 or the environmental performance where not specified in Sections 4 and 5;
- (c) the environmental performance of combinations of techniques.

All underlying data, required by Directive 2003/4/EC of the European Parliament and of the Council⁸, methodologies, and assumptions, must be fully documented and publicly available.

⁸

Directive 2003/4/EC of the European Parliament and of the Council of 28 January 2003 on public access to environmental information and repealing Council Directive 90/313/EEC (OJ L 41, 14.2.2003, p. 26-32, ELI <http://data.europa.eu/eli/dir/2003/4/oj>)

1. UNIFORM CONDITIONS FOR PIG REARING

The UC in this section apply in addition to the general Uniform Conditions listed in Section 3.

1.1. Nutritional management in pig farms

UC 1. Diet formulations and nutritional strategies shall be used, to minimise total nitrogen and total phosphorus excreted, thus preventing and/or minimising nitrogen and phosphorus emissions and maximising nitrogen and phosphorus efficiency, while meeting the nutritional needs of the animals.

Total nitrogen and total phosphorus excreted shall not exceed the values listed in Table 1.1 and Table 1.2.

Table 1.1: Maximum values of total nitrogen excreted, by pig category

Parameter	Pig category	Maximum values of total nitrogen excreted (kg N excreted / animal place / year) ⁽¹⁾ ⁽²⁾
Total nitrogen excreted, expressed as N	Weaners	3,6
	Fattening pigs	12,0
	Sows (including piglets)	25,0
⁽¹⁾ For Iberian pigs, the maximum values may be up to 30 % higher. ⁽²⁾ For heavy pigs other than Iberian pigs, the maximum values may be up to 20 % higher.		

Member States may apply maximum values of total nitrogen excreted across all pig categories at the same farm, subject to no net increase in nitrogen excretion in the farm.

Table 1.2: Maximum values of total phosphorus excreted, by pig category

Parameter	Pig category	Maximum values of total phosphorus excreted (kg P ₂ O ₅ excreted / animal place / year) ⁽¹⁾
Total phosphorus excreted, expressed as P ₂ O ₅	Weaners	1,5
	Fattening pigs	4,5
	Sows (including piglets)	12,0
⁽¹⁾ In the case of co-products (e.g. whey) used as feed as part of a documented resource-efficient nutritional strategy, or to comply with binding feed specifications for recognised quality schemes, the maximum values may be up to 20 % higher.		

Member States may apply maximum values of total phosphorus excreted across all pig categories at the same farm, subject to no net increase in phosphorus excretion in the farm.

The associated monitoring is provided for in Section 3.7.1.

Values equal to or below those set out in Table 1.1 and Table 1.2 can be achieved by using one or a combination of the techniques listed below.

Technique ⁽¹⁾		Parameter(s)	Applicability
a	Limitation of the crude protein content in the diet by using a N-balanced diet based on the energy needs and digestible amino acids	Total nitrogen	Generally applicable.
b	Phase feeding with a diet formulation adapted to the specific requirements of the production period	Total nitrogen, total phosphorus	Generally applicable.
c	Controlled supply of amino acids in crude-protein-reduced diets	Total nitrogen	Generally applicable within the constraints associated with the market availability of low-protein feedstuff and/or synthetic amino acids.
d	Use of authorised feed additives which reduce the total nitrogen excreted / ammonia emissions	Total nitrogen	Generally applicable.
e	Use of authorised feed additives which reduce the total phosphorus excreted (e.g. phytase)	Total phosphorus	Generally applicable.
f	Use of highly digestible inorganic phosphates to partially replace conventional sources of phosphorus in the feed	Total phosphorus	Generally applicable within the constraints associated with the market availability of highly digestible inorganic phosphates.
g	Precision feeding in pig rearing for individual animals	Total nitrogen, total phosphorus	Applicability to existing plants may be restricted by building and other structural constraints.
⁽¹⁾ A description of the techniques is provided for in Section 4.1.			

1.2. Ammonia and methane emissions from animal houses in pig farms

UC 2. Animal housing shall be designed and operated with techniques to prevent or, where that is not practicable, minimise ammonia and methane emissions to air. Ammonia emissions to air from each animal house shall not exceed the values set out in Table 1.3.

Table 1.3: Maximum values of ammonia emissions to air, by pig category

Substance	Pig category	Maximum values of ammonia emissions to air (kg NH ₃ / animal place / year) ⁽¹⁾ ⁽²⁾		
		New plants (other than specific plants)	Existing plants (other than specific plants)	Specific plants ⁽³⁾ ⁽⁴⁾
Ammonia, expressed as NH ₃	Weaners	0,4	0,5 ⁽⁶⁾	0,7
	Fattening pigs	2,0	2,4 ⁽⁷⁾	3,5
	Mating and gestating sows ⁽⁵⁾	2,3	2,7	4,0
	Farrowing sows (including piglets) ⁽⁵⁾	3,6	4,2 ⁽⁸⁾	7,0
⁽¹⁾ For Iberian pigs, the maximum values may be up to 30 % higher. ⁽²⁾ For heavy pigs other than Iberian pigs, the maximum values may be up to 20 % higher. ⁽³⁾ New or existing plants with solid manure systems. ⁽⁴⁾ Existing plants with a deep pit (fully or partly slatted floor) combined only with nutritional management. ⁽⁵⁾ Maximum values may not apply to systems enhancing animal welfare (e.g. loose housing systems for farrowing sows) that are recognised at Union, national or international level. ⁽⁶⁾ For newly covered farms, the maximum values may be higher and up to 0,7 kg NH ₃ / animal place / year. ⁽⁷⁾ For newly covered farms, the maximum values may be higher and up to 2,6 kg NH ₃ / animal place / year. ⁽⁸⁾ For newly covered farms, the maximum values may be higher and up to 5,6 kg NH ₃ / animal place / year.				

As an alternative to the maximum values at each animal house, Member States may apply maximum values across all animal houses at the same farm, subject to no net increase in ammonia emissions from housing in the farm.

The associated monitoring is provided for in Section 3.7.2.

The uniform condition UC 2 and values equal to or below those set out in Table 1.3 can be achieved by using UC 1 and one or a combination of the techniques listed below.

Technique ⁽¹⁾		Applicability
a	Deep pit (fully or partly slatted floor) combined with: (a) at least nutritional management techniques in existing plants; and (b) additional mitigation measures (e.g. air cleaning system, slurry acidification, slurry cooling, use of slurry additives, frequent slurry removal) in new plants.	Generally applicable.
b	Frequent slurry removal, e.g. by vacuum system, scraper, flushing (in the case of a fully or partly slatted floor)	Applicability to existing plants may be restricted by building and other structural constraints. Flushing may not be applicable to plants located close to sensitive receptors if this causes odour peaks.
c	Slanted walls or funnel system (in the case of a fully or partly slatted floor)	Applicability to existing plants may be restricted by building and other structural constraints.

Technique ⁽¹⁾		Applicability
d	Reduced manure pit (in the case of a partly slatted floor)	Applicability to existing plants may be restricted by building and other structural constraints. Only applicable to fattening pigs and mating and gestating sows.
e	Full litter system (in the case of a solid concrete floor)	Full litter systems and kennels are only applicable to mating and gestating sows, weaners and fattening pigs. Straw flow systems are only applicable to weaners, fattening pigs and gestating sows. May not be applicable to naturally ventilated plants located in warm climates and to existing plants with forced ventilation for weaners and fattening pigs. Kennels may not be applicable to existing plants due to a lack of space.
f	Kennel / hut housing (in the case of a partly slatted floor)	
g	Straw flow system (in the case of a solid concrete floor)	Only applicable to weaners and fattening pigs. May not be applicable to existing plants due to technical constraints.
h	Convex floor and separated manure and water channels (in the case of partly slatted pens)	Only applicable to farrowing sows. May not be applicable to existing plants due to technical constraints.
i	Littered pens with combined manure generation (slurry and solid manure)	Only applicable to mating and gestating sows. Not applicable to existing plants without solid concrete floors.
j	Feeding/lying boxes on solid floor (in the case of litter pens)	Only applicable to farrowing sows.
k	Manure pan (in the case of a fully or partly slatted floor)	Only applicable to weaners, fattening pigs and mating and gestating sows. May not be applicable to existing plants due to technical constraints.
l	Manure collection in water	Only applicable to fattening pigs. Applicability to existing plants may be restricted by building and other structural constraints.
m	V-shaped manure belts (in the case of a partly slatted floor)	Only applicable to farrowing sows. Applicability to existing plants may be restricted by building and other structural constraints.
n	A combination of water and manure channels (in the case of a fully slatted floor)	Only applicable to fattening pigs. Not applicable to cold climates. Applicability to existing plants may be restricted by building and other structural constraints.
o	Littered external alley (in the case of a solid concrete floor)	Applicable to existing plants only where a centralised ventilation system is used.
p	Use of an air cleaning system, such as: (a) wet acid scrubber; (b) two-stage or three-stage air cleaning system; (c) bioscrubber (or biotrickling filter).	Only applicable to fattening pigs. Not applicable to plants equipped with pits that have slanted walls and to plants that apply slurry removal by flushing.
q	Use of floating balls in the manure channel	

Technique ⁽¹⁾		Applicability
r	Slurry cooling	Not applicable when heat reuse is not possible and when litter is used. Slurry cooling with fins may not be applicable in the case of high dry matter content.
s	Slurry acidification	May not be applicable in the case of structural constraints (e.g. unsuitable concrete quality) and in the event that the slurry handling and storage system lacks acid-resistant coating.
t	Housing with distinct living zones for animals	Applicability to existing plants may be restricted by building and other structural constraints.
u	Use of the following additives for the treatment of manure: (a) ion-exchange-based materials; (b) selected microorganisms.	Applicability of the use of microorganisms may be restricted by health and safety issues.
v	Gas-permeable membranes (GPM) for ammonia capture from gaseous media	May not be applicable in existing plants due to a lack of space.
⁽¹⁾ A description of the techniques is provided for in Sections 4.2.1 and 4.2.3.		

As an alternative to the maximum values set out in Table 1.3, Member States may use a relative system based on minimum ammonia reduction requirements at each animal house as compared to a reference system based on Union or international sources (e.g. UNECE guidance document on ‘Options for ammonia mitigation’), provided that they ensure at least the same reduction of ammonia emissions and that they are based on scientifically established methods.

In addition, the emerging techniques in Section 5.1 have been identified.

2. UNIFORM CONDITIONS FOR POULTRY REARING

The Uniform Conditions in this section apply in addition to the general Uniform Conditions given in Section 3.

2.1. Nutritional management in poultry farms

UC 3. For all poultry categories, diet formulations and nutritional strategies shall be used, to minimise total nitrogen and total phosphorus excreted, thus preventing and/or minimising nitrogen and phosphorus emissions and maximising nitrogen and phosphorus efficiency, while meeting the nutritional needs of the animals.

For laying hens, broilers, ducks and turkeys, total nitrogen and total phosphorus excreted shall not exceed the values given in Table 2.1 and Table 2.2.

Table 2.1: Maximum values of total nitrogen excreted, by poultry category

Parameter	Poultry category	Maximum values of total nitrogen excreted (kg N excreted / animal place / year) ⁽¹⁾
Total nitrogen excreted, expressed as N	Laying hens	0,80
	Broilers	0,55
	Ducks	0,80
	Male turkeys	2,00
	Mixed male and female turkeys	1,70
	Female turkeys	1,20
⁽¹⁾ The maximum values do not apply to organic production.		

Member States may apply maximum values of total nitrogen excreted across all poultry categories at the same farm, subject to no net increase in nitrogen excretion in the farm.

Table 2.2: Maximum values of total phosphorus excreted, by poultry category

Parameter	Poultry category	Maximum values of total phosphorus excreted (kg P ₂ O ₅ excreted / animal place / year) ⁽¹⁾
Total phosphorus excreted, expressed as P ₂ O ₅	Laying hens	0,40
	Broilers	0,20
	Ducks	0,30
	Male turkeys	1,00
	Mixed male and female turkeys	0,80

	Female turkeys	0,50
⁽¹⁾ The maximum values do not apply to organic production.		

Member States may apply maximum values of total phosphorus excreted across all poultry categories at the same farm, subject to no net increase in phosphorus excretion in the farm.

The associated monitoring is given in Section 3.7.1.

The uniform condition UC 3 and values equal to or below those listed in Table 2.1 and Table 2.2 can be achieved by using one or a combination of the techniques given below.

Technique ⁽¹⁾		Parameter(s)	Applicability
a	Limitation of the crude protein content in the diet by using a N-balanced diet based on the energy needs and digestible amino acids	Total nitrogen	Generally applicable.
b	Phase feeding with a diet formulation adapted to the specific requirements of the production period	Total nitrogen, Total phosphorus	Generally applicable.
c	Controlled supply of amino acids in crude-protein-reduced diets	Total nitrogen	Generally applicable within the constraints associated with the market availability of low-protein feedstuff and/or synthetic amino acids. Synthetic amino acids are not applicable to organic production.
d	Use of authorised feed additives which reduce the total nitrogen excreted/ ammonia emissions	Total nitrogen	Generally applicable.
e	Use of authorised feed additives which reduce the total phosphorus excreted (e.g. phytase)	Total phosphorus	The use of phytase may not be applicable in case of organic production.
f	Use of highly digestible inorganic phosphates to partially replace conventional sources of phosphorus in the feed	Total phosphorus	Generally applicable within the constraints associated with the market availability of highly digestible inorganic phosphates.
⁽¹⁾ A description of the techniques is given in Section 4.1.			

2.2. Ammonia emissions from animal houses in poultry farms

UC 4. For all poultry categories, animal housing shall be designed and operated with techniques to prevent or, where that is not practicable, minimise ammonia emissions to air. For laying hens, broilers, ducks and turkeys, ammonia emissions to air from each animal house shall not exceed the values given in Table 2.3.

Table 2.3: Maximum values of ammonia emissions to air, by poultry category

Substance	Poultry category	Maximum values of ammonia emission to air (kg NH ₃ / animal place / year) ⁽¹⁾	
		New plants	Existing plants
Ammonia, expressed as NH ₃	Laying hens (in cage systems)	0,04	0,08
	Laying hens (in non-cage systems)	0,10	0,13 ⁽²⁾
	Broilers	0,05	0,07
	Ducks	0,17	
	Male turkeys	0,55 ⁽³⁾	
	Mixed male and female turkeys	0,44 ⁽⁴⁾	
	Female turkeys	0,35	

⁽¹⁾ The maximum values do not apply to organic production.

⁽²⁾ For existing plants with infrequent manure removal (in case of deep litter with a manure pit), in combination with a measure achieving a high dry matter content of the manure, the maximum value may be higher and up to 0,25 kg NH₃/animal place/year.

⁽³⁾ For existing plants for which only nutritional management techniques can be used, the maximum value may be higher and up to 0,61 kg NH₃/animal place/year.

⁽⁴⁾ The maximum value for fattening ‘mixed male and female turkeys’ may be higher and shall take into account the proportion of males and females as well as the respective fattening periods.

As an alternative to the maximum values at each animal house, Member States may apply maximum values across all animal houses at the same farm, subject to no net increase in ammonia emissions from housing in the farm.

The associated monitoring is listed in Section 3.7.2.

The uniform condition UC 4 and values equal to or below those set out in Table 2.3 can be achieved by using UC 3 and one or a combination of the techniques listed below.

Technique ⁽¹⁾		Applicability
All poultry species		
a	Use of an air cleaning system, such as: (a) wet acid scrubber; (b) two-stage or three-stage air cleaning system; (c) bioscrubber (or biotrickling filter).	Applicable to existing plants only where a centralised ventilation system is used.

Technique (¹)		Applicability
b	Gas-permeable membranes (GPM) for ammonia capture from gaseous media.	May not be applicable in existing plants due to a lack of space.
Laying hens, broiler breeders, pullets (cage systems)		
c	Manure removal by belts with at least: (a) one removal per week with air drying; or (b) two removals per week without air drying.	Generally applicable.
Laying hens, broiler breeders, pullets (non-cage systems)		
d	Forced or natural ventilation system without frequent manure removal in existing plants with deep litter and a manure pit, combined with at least an additional mitigation measure (e.g. achieving a high dry matter content of the manure, air cleaning system).	Generally applicable.
e	Manure belt or scraper (in the case of deep litter with a manure pit).	Applicability to existing plants may be restricted by building and other structural constraints.
f	Forced air drying of manure via tubes (in the case of deep litter with a manure pit).	Only applicable to plants with sufficient space underneath the slats.
g	Forced air drying of manure using a perforated floor (in the case of deep litter with a manure pit).	Applicability to existing plants may be restricted by building and other structural constraints.
h	Drying of litter on manure belt (in the case of aviaries).	May not be applicable to existing plants in the event of insufficient width of the shed.
i	Forced air drying of litter on manure belt (in the case of aviaries).	Generally applicable.
j	Forced drying of litter using indoor air (in the case of a solid floor with deep litter).	Generally applicable.
k	Litter slide (in the case of aviaries).	Only applicable to laying hens.
Broilers		
l	Forced drying of litter using indoor air (in the case of a solid floor with deep litter).	May not be applicable to existing plants in the event of insufficient ceiling height. May not be applicable to warm climates, depending on the indoor temperature.
m	Forced drying of litter using heated indoor air (in the case of a solid floor with deep litter).	May not be applicable to warm climates, depending on the indoor temperature. May not be applicable to existing plants in the event of insufficient ceiling height.
n	Natural or forced ventilation with a non-leaking drinking system (in the case of a solid floor with deep litter).	Natural ventilation is not applicable to plants with a centralised ventilation system. Natural ventilation may not be applicable during the initial stage of rearing or due to extreme climate conditions.
o	Heated and cooled littered floor (in the case of combideck systems).	For existing plants, the applicability depends on the possibility to install closed underground storage for the circulating water. Applicability may be restricted by animal welfare considerations.
p	Litter composed of straw and sodium hydrogen sulphate.	Generally applicable.
Ducks		

Technique ⁽¹⁾		Applicability
q	Frequent litter addition (in the case of a solid floor with deep litter or deep litter combined with a slatted floor).	For existing plants with deep litter combined with a slatted floor, the applicability may be restricted by the design of the existing structure.
r	Frequent manure removal (in the case of a fully slatted floor).	Only applicable to the rearing of Barbary/Muscovy ducks (<i>Cairina Moschata</i>), for sanitary reasons.
Turkeys		
s	Natural or forced ventilation with a non-leaking drinking system (in the case of a solid floor with deep litter).	Natural ventilation is not applicable to plants with a centralised ventilation system. Natural ventilation may not be applicable during the initial stage of rearing or due to extreme climate conditions.
t	Forced drying of litter using heated indoor air (in the case of a solid floor with deep litter)	May not be applicable to warm climates, depending on the indoor temperature. May not be applicable to existing plants in the event of insufficient ceiling height.
⁽¹⁾ A description of the techniques is given in Sections 4.2.2 and 4.2.3.		

As an alternative to the maximum values given in Table 2.3, Member States may use a relative system based on minimum ammonia reduction requirements at each animal house as compared to a reference system based on European or international sources (e.g. UNECE guidance document on ‘Options for ammonia mitigation’), provided that they ensure at least the same reduction of ammonia emissions and that they are based on scientifically established methods.

In addition, the emerging techniques in Section 5.1 have been identified.

3. GENERAL UNIFORM CONDITIONS (FOR PIG AND POULTRY REARING)

The Uniform Conditions in this section apply in addition to the Uniform Conditions given in Section 1 (for pig rearing) and in Section 2 (for poultry rearing).

3.1. Housing systems

UC 5. Animal houses shall be designed and operated with techniques to:

- i. prevent or, where that is not practicable, minimise emissions to air of:
 - (1) ammonia and methane (see Section 1.2 for pig houses and Section 2.2 for poultry houses);
 - (2) dust; this can be achieved by using one or a combination of the techniques (a), (b) and (c) given below;
 - (3) odour; this can be achieved by using one or a combination of the techniques (d), (e) and (f) given below; for odour, see also Section 3.6.1;
- ii. use water efficiently; this can be achieved by using a combination of the techniques (g), (h) and (i) given below; for the efficient use of water, see also Section 3.6.3;
- iii. use energy efficiently; this can be achieved by using one or a combination of the techniques (j) to (p) given below; for energy, see also Section 3.6.4.

Animal houses and their structures and equipment shall be regularly checked and maintained.

Technique		Description	Applicability
Ammonia and methane emissions			
See Section 1.2 for ammonia and methane emissions in pig houses and Section 2.2 for ammonia emissions in poultry houses.			
Dust emissions			
a	Prevention or reduction of dust generation inside livestock buildings by using, e.g. low-dust littering techniques, coarser litter material, litter slides, <i>ad libitum</i> feeding, low-dust feed, dust separators, optimised airspeed ventilation	See Section 4.2.4	For the use of coarser litter material, long straw is only applicable to litter-based housing systems. For <i>ad libitum</i> feeding, applicability may be limited to certain feed delivery systems (e.g. dry feeding systems). For optimised airspeed ventilation, applicability may be restricted by animal welfare considerations.
b	Reduction of dust concentration inside livestock buildings by applying, e.g. water fogging, ionisation, oil spraying		For water fogging, applicability may be restricted by animal welfare considerations. Applicability may also be restricted for solid manure systems at the end of the rearing period due to high ammonia emissions. Ionisation may not be applicable to pig plants or to existing poultry plants due to technical constraints (e.g. electrical installation). For oil spraying, applicability may be

Technique		Description	Applicability
			restricted by animal welfare considerations. This technique is only applicable to poultry plants with birds older than around 21 days. The applicability to plants for laying hens may be limited due to the risk of contamination of the equipment present in the shed.
c	Dust abatement using an air cleaning system, e.g. dry filter, two-stage or three-stage air cleaning system, water scrubber, water trap, wet acid scrubber, biofilter, bioscrubber		Water scrubbers, wet acid scrubbers and bioscrubbers are applicable to existing plants only where a centralised ventilation system is used. Biofilters are only applicable to slurry-based plants and straw-based systems. Water traps are only applicable to plants with a tunnel ventilation system. Dry filters are only applicable to poultry plants with a tunnel ventilation system.
Odour emissions			
d	Odour prevention and reduction measures inside the housing systems, e.g. keeping animals and surfaces dry and clean, reducing the manure-emitting surface, removing manure frequently, reducing the temperature of manure and the indoor environment, decreasing air flow	See Section 4.2.4	Techniques reducing air temperature and/or flow may not be applicable due to animal welfare considerations. For frequent manure removal, flushing may not be applicable to plants located close to sensitive receptors if this causes odour peaks.
e	Optimisation of discharge conditions of exhaust air from the animal houses, e.g. increasing the outlet height and/or ventilation velocity		Generally applicable.
f	Odour abatement using an air cleaning system, e.g. bioscrubber (or biotrickling filter), biofilter, two-stage or three-stage air cleaning system)		Applicable to existing plants only where a centralised ventilation system is used.
Water consumption			
g	Selection and use of suitable equipment (e.g. nipple drinkers, round drinkers, water troughs) for the specific animal category while ensuring water availability (<i>ad libitum</i>)	See Section 4.2.4	Generally applicable.
h	Verification and (if necessary) adjustment, on a regular basis, of the calibration of the drinking water equipment		Generally applicable.
i	Use of techniques such as mechanical dry cleaning or high-pressure cleaners for animal housing and equipment		High pressure cleaners are not applicable to poultry plants using dry cleaning systems.
Energy consumption			
j	High-efficiency heating/cooling and ventilation systems	See Section 4.2.4	Applicability to existing plants may be restricted by building and other structural constraints.
k	Optimisation of heating/cooling and ventilation systems and management, especially where air cleaning systems are used		Generally applicable.
l	Insulation of the walls, floors and/or		May not be applicable to plants using

Technique		Description	Applicability
	ceilings of animal housing		natural ventilation. Applicability to existing plants may be restricted by building and other structural constraints.
m	Heat recovery (e.g. combideck system, heat recovery ventilation systems, slurry cooling heat recovery)		Applicability depends on the possibility to install closed underground storage for the circulating water.
n	Use of natural ventilation		Not applicable to plants with a centralised ventilation system. In pig plants, this technique may not be applicable to: (a) housing systems with littered floors in warm climates; (b) housing systems without littered floors or without covered, insulated boxes (e.g. kennels) in cold climates. In poultry plants, this technique may not be applicable: (a) during the initial stage of rearing, apart from duck production; (b) in the event of extreme climate conditions.
o	Use of hybrid ventilation		In pig plants, this technique may not be applicable to housing systems with littered floors in warm climates.
p	Evaporative cooling systems using pads and water recirculation		Only applicable to housing systems requiring temperature control.

3.2. Manure storage

UC 6. Manure storage shall be carried out with techniques to prevent or, where that is not practicable, minimise emissions to air of ammonia, methane and odour, and emissions to soil and water of nitrogen, phosphorus and microbial pathogens:

- i. for storage of slurry in stores, this can be achieved by using techniques (a) to (e) and a combination of the techniques (f) to (j) listed below;**
- ii. for storage of slurry in earth-banked storage, this can be achieved by using techniques (a) to (e), (k), and one or both of the techniques (f) and (g) listed below;**
- iii. for storage of solid manure, this can be achieved by using technique (a) and one of the techniques (l), (m) and (n) given below;**
- iv. for temporary solid manure heaps, this can be achieved by using techniques (o) (p) and (q) given below;**

For odour, see also Section 3.6.1.

Manure storage facilities shall be regularly checked and maintained.

Technique ⁽¹⁾		Main environmental issue(s) targeted	Applicability
All types of manure			
a	Use of a storage facility with a sufficient capacity to hold the manure during periods in which landspreading is not possible	Nitrogen, phosphorus, microbial pathogens, ammonia, odour	Generally applicable.
Slurry – general techniques			
b	Use of storage facilities that are able to withstand mechanical, chemical and thermal influences	Nitrogen, phosphorus, microbial pathogens	Generally applicable.
c	Use of leakproof facilities and equipment for collection and transfer of slurry (e.g. pits, channels, drains, pump stations)	Nitrogen, phosphorus, microbial pathogens	Generally applicable.
d	Verification of the structural integrity of storage facilities at least once every year	Ammonia, methane, odour, nitrogen, phosphorus, microbial pathogens	May not be applicable to underground slurry storage.
e	Coverage ensuring the highest environmental performance, with one of the following techniques: (a) rigid covers; (b) flexible covers; (c) floating covers.	Ammonia, odour	Rigid covers are not applicable to earth-banked storage and may not be applicable to existing plants due to structural limitations to withstand the extra load. Flexible covers are not applicable to areas where prevailing weather conditions can compromise their structure. For floating covers:

Technique ⁽¹⁾		Main environmental issue(s) targeted	Applicability
			- applicability may be restricted by technical constraints (e.g. in the event that floating materials compromise the correct functioning of the pumps); (a) the use of pellets, light bulk materials and geometrical plastic tiles is not applicable to naturally crusting slurries; (b) natural crust formation may not be applicable to cold climates, to slurry with a low dry matter content and in the event that stirring, filling and/or discharging of slurry renders the natural crust unstable; (c) the use of straw and light bulk materials may not be applicable to large earth-banked storage spaces where wind drift does not permit the surface to be kept fully covered.
f	Installation of a leakage detection system	Nitrogen, phosphorus, microbial pathogens	Only applicable to new plants.
g	Minimisation of stirring of slurry	Ammonia, odour	Generally applicable.
Slurry – techniques applying to slurry stores			
h	Appropriate design of the slurry store by using techniques such as: (a) reduction of the ratio between the emitting surface area and the volume of the slurry store; (b) reduction of wind velocity and air exchange on the slurry surface by operating the store at a lower level of fill.	Ammonia, odour	The reduction of the ratio between emitting surface area and volume may not be applicable to existing slurry stores.
i	Slurry acidification	Ammonia, methane	May not be applicable in the case of structural constraints (e.g. unsuitable concrete quality) and in the event that the storage system lacks acid-resistant coating. May not be applicable to stores with rigid covers.
j	Slurry cooling	Ammonia, methane	May not be applicable to stores with rigid covers.
Slurry – technique applying to earth-banked storage			
k	Impermeable base and walls, e.g. with plastic lining (or double-lined), and leakproof structures	Nitrogen, phosphorus, microbial pathogens	Generally applicable.
Storage of solid manure			
l	Storage of dried solid manure in a barn	Ammonia, methane, nitrogen,	Generally applicable.

Technique ⁽¹⁾		Main environmental issue(s) targeted	Applicability
		phosphorus, microbial pathogens	
m	Use of a concrete silo for storage of solid manure	Nitrogen, phosphorus, microbial pathogens	Generally applicable.
n	Storage of solid manure on a solid impermeable floor designed to prevent leaks to the environment	Nitrogen, phosphorus, microbial pathogens	Generally applicable.
Temporary solid manure heaps			
o	Appropriate distances between temporary solid manure heaps and water bodies, with annual change of location and minimisation of time on field	Nitrogen, phosphorus, microbial pathogens	Generally applicable.
p	Reduction of the ratio between the emitting surface area and the volume of the temporary solid manure heap	Ammonia, odour	Generally applicable.
q	Coverage of temporary solid manure heaps	Ammonia, odour, nitrogen, phosphorus, microbial pathogens	Generally applicable when solid manure is dried or pre-dried in animal housing. May not be applicable to undried solid manure in the event of frequent addition to the heap.
⁽¹⁾ A description of the techniques is given in Section 4.3.			

In addition, the emerging technique in Section 5.2 has been identified.

3.3. On-farm processing of manure

UC 7. If on-farm processing of manure is used, this shall be carried out with techniques to prevent or, where that is not practicable, minimise emissions of nitrogen, phosphorus, methane, odour and microbial pathogens to the environment, while facilitating manure management, including storage and/or landspreading. This can be achieved by using one or a combination of the techniques given below.

	Technique ⁽¹⁾	Applicability
a	Mechanical separation of slurry, e.g. using: (a) screw press separator; (b) decanter-centrifuge separator; (c) separation by sieves; (d) filter pressing.	The use of polyacrylamide as a flocculant may not be applicable due to the risk of acrylamide formation.
b	Anaerobic digestion	May not be applicable in the case of an insufficient mass of manure available on the farm.
c	Manure drying with an external tunnel	Only applicable to manure from plants for laying hens. Not applicable to existing plants without manure belts.

	Technique ⁽¹⁾	Applicability
d	Mechanical separation of slurry into solid and liquid fractions, followed by aerobic digestion (aeration) of the liquid fraction, with nutrient recovery	Applicability may be restricted in cold climates, due to the difficulty to maintain the required level of aeration during winter.
e	Mechanical separation of slurry into solid and liquid fractions, followed by nitrification-denitrification of the liquid fraction, with nutrient recovery	Generally applicable.
f	Composting of solid manure	May not be applicable to existing farms due to a lack of space in the event that windrows are used for composting.
g	Vacuum evaporation with ammonia recovery	May not be applicable to existing farms if waste heat from other processes is not available.
h	Vacuum fine separators with screens	Generally applicable.
i	Pressure and flocculation solid-liquid separators	Generally applicable.
j	Mechanical separation of slurry in a solid and liquid fraction, followed by treatment in constructed wetlands of the liquid fraction, with nutrient recovery	Applicability may be restricted in cold climates. May not be applicable in existing plants due to a lack of space.
⁽¹⁾ A description of the techniques is given in Section 4.4.		

In addition, the emerging techniques in Section 5.3 have been identified.

3.4. On-farm manure landspreading

UC 8. In the case of on-farm landspreading of manure, this shall be carried out with techniques to:

- i. prevent or, where that is not practicable, minimise emissions of nitrogen, phosphorus and microbial pathogens to soil and water; this can be achieved by using all of the techniques (a), (b) and (c) and technique (i) given below;**
- ii. prevent or, where that is not practicable, minimise ammonia and odour emissions to air; this can be achieved by using one or a combination of the techniques (d) to (h) given below, in combination with technique (i); for odour, see also Section 3.6.1.**

Techniques with the highest environmental performance shall be used, if applicable.

	Technique ⁽¹⁾	Main environmental issue(s) targeted	Applicability
a	Sufficient distance between landspreading areas and areas at risk of leaching/run-off, as well as neighbouring areas	Nitrogen, phosphorus, microbial pathogens	Generally applicable.
b	Application restrictions during run-off risk conditions	Nitrogen, phosphorus, microbial pathogens	Generally applicable.
c	Landspreading taking into account crop nutrient requirements	Nitrogen, phosphorus	Generally applicable.

Technique ⁽¹⁾		Main environmental issue(s) targeted	Applicability
d	Slurry acidification prior to application	Ammonia	Generally applicable.
e	Use of diluted slurry in low-pressure water irrigation systems	Ammonia	Not applicable to crops grown to be eaten raw due to the risk of contamination. Not applicable when the soil type does not allow rapid infiltration of diluted slurry into the soil. Not applicable when crops do not require irrigation.
f	Slurry application through band spreaders, using one of the following techniques: trailing hose; trailing shoe	Ammonia, odour	Applicability may be limited when the straw content of the slurry is too high or when the dry matter content of the slurry is higher than 10 %. Trailing shoe is not applicable when growing solid-seeded arable crops.
g	Slurry application through shallow injectors (open slot)	Ammonia, odour	Not applicable on stony, shallow or compacted soils where it is difficult to achieve a uniform penetration. Applicability may be limited where crops may be damaged by machinery.
h	Slurry application through deep injectors (closed slot)	Ammonia, odour	Not applicable on stony, shallow or compacted soil where it is difficult to achieve a uniform penetration and an effective slit closure. Not applicable during the vegetation of the crops. Not applicable on grassland, unless changing to arable land or when reseeding.
i	Manure incorporation in the soil as soon as possible after application (with a time delay up to four hours or up to 12 hours when conditions are not favourable for a faster incorporation)	Ammonia, odour	Not applicable to grassland and conservation tillage, unless changing to arable land or when reseeding. Not applicable to cultivated land with crops that can be damaged. Not applicable to slurry applied using shallow or deep injectors.
⁽¹⁾ A description of the techniques is given in Section 4.5.			

3.5. On-farm storage of dead animals

UC 9. On-farm storage of dead animals shall be carried out preventing or, where that is not practicable, minimising emissions of odour. This can be achieved through the use of authorised storage systems in line with relevant legislation to ensure public and animal health protection.

Description: See Section 4.6.

Applicability: Applicable to fallen stock that is not immediately collected by an authorised collector or treated on the farm.

3.6. Farm-wide activities

3.6.1. Odour emissions

UC 10. Odour emissions and/or odour impact shall be prevented or, where that is not practicable, minimised, ensuring appropriate distances between the plant/farm and sensitive receptors, and by using an appropriate combination of techniques in animal housing (see Section 3.1), on-farm manure storage (see Section 3.2), on-farm manure processing (see Section 3.3) and on-farm manure landspreading (see Section 3.4). In case odour nuisance at sensitive receptors has been substantiated, follow-up actions shall be implemented, e.g. an odour management plan.

Description: See Section 4.7.1.

Applicability: Ensuring appropriate distances between the plant/farm and sensitive receptors is only applicable to new farms/plants.

3.6.2. Noise emissions

UC 11. Noise emissions and/or noise impact shall be prevented or, where that is not practicable, minimised, using one or a combination of the techniques given below.

Applicability: This Uniform Condition is only applicable to cases where a noise nuisance at sensitive receptors is expected and/or has been substantiated.

Technique ⁽¹⁾		Applicability
a	Appropriate distances between the farm/plant and sensitive receptors	Only applicable to new farms/plants.
b	Appropriate equipment location	In the case of existing plants, the relocation of equipment may be restricted by a lack of space.
c	Operational measures	Generally applicable.
d	Low-noise equipment	Feeding systems which reduce the pre-feeding stimulus are only applicable to pig plants. Passive <i>ad libitum</i> feeders are only applicable when the equipment is new or replaced or when animals do not require restricted feeding.
e	Noise-control equipment	Applicability may be restricted due to space requirements, and health and safety issues. May not be applicable if noise-absorbent materials impede the effective cleaning of the plant.
f	Noise abatement techniques	Applicability may be restricted by biosecurity reasons.
⁽¹⁾ A description of the techniques is given in Section 4.7.2.		

3.6.3. Water management

UC 12. Water shall be used efficiently and waste water properly managed. This can be achieved by using technique (a) and one or a combination of the techniques (b) to (g) given below. For the efficient use of water in housing systems, see Section 3.1.

Technique ⁽¹⁾		Applicability
a	Detection and repair of water leaks	Generally applicable.
b	Drainage of waste water to a dedicated container or to a slurry store	Generally applicable when waste water is drained to a dedicated container. Applicability may be restricted when waste water is drained to a slurry store due to biosecurity risks and/or the waste water's chemical characteristics.
c	Segregation of uncontaminated rainwater from waste water streams that require treatment	May not be applicable to existing farms.
d	Reuse of uncontaminated rainwater as cleaning water	May not be applicable to existing farms. Applicability may be restricted by biosecurity risks and/or the rainwater's chemical characteristics.
e	Keeping the fouled yard areas as small as possible	Generally applicable.
f	Treatment of waste water	Generally applicable.
g	Use of waste water in irrigation and landspreading	Applicability may be restricted due to the limited availability of suitable land adjacent to the farm and/or waste water chemical characteristics. Applicability may be restricted by legislative requirements on the use of waste water (e.g. Regulation (EU) 2020/741 ⁹ of the European Parliament and of the Council on minimum requirements for water reuse).
⁽¹⁾ A description of the techniques is given in Section 4.7.3.		

3.6.4. Energy management

UC 13. Energy shall be used efficiently. This can be achieved by using one or a combination of the techniques given below. For the efficient use of energy in housing systems, see Section 3.1.

	Technique ⁽¹⁾	Applicability
a	Use of energy-efficient lighting	Generally applicable.
b	Use of heat exchangers. One of the following systems may be used: (a) air-air; (b) air-water; (c) air-ground.	Air-ground heat exchangers are only applicable when there is available space due to the need for a large soil surface.
c	Use of heat pumps for heat recovery	The applicability of heat pumps based on geothermal heat recovery is limited when using horizontal pipes due to the need for space availability.
d	Renewable energy generation	The applicability of a specific renewable energy generation may be limited due to climate conditions. May not be applicable to existing farms due to building and/or space constraints.
e	Insulation of the walls, floors and/or	Applicability to existing plants may be restricted by

⁹

Regulation (EU) 2020/741 of the European Parliament and of the Council of 25 May 2020 on minimum requirements for water reuse (OJ L 177, 5.6.2020, p. 32, ELI <http://data.europa.eu/eli/reg/2020/741/oj>).

	Technique ⁽¹⁾	Applicability
	ceilings of buildings/structures	building and other structural constraints.
⁽¹⁾ A description of the techniques is given in Section 4.7.4.		

3.7. Monitoring

3.7.1. Monitoring of nitrogen and phosphorus excretion

UC 14. Total nitrogen and total phosphorus excreted in manure shall be monitored using the technique given below.

	Technique ⁽¹⁾	Frequency	Applicability
a	Mass balance of nitrogen and phosphorus based on the feed intake, crude protein and total phosphorus content in the diet, and animal performance ⁽²⁾	Once every year for each animal category ⁽³⁾	Generally applicable.
⁽¹⁾ A description of the technique is given in Section 4.8.1. ⁽²⁾ The monitoring technique is further developed at Member State level. ⁽³⁾ If feed intake, crude protein and total phosphorus content in the diet, and animal performance are proven to be sufficiently stable, the monitoring frequency may be reduced to every time significant changes arise.			

3.7.2. Monitoring of ammonia emissions to air

UC 15. Ammonia emissions to air shall be monitored at least from each animal house using at least one of the techniques given below.

	Technique ⁽¹⁾	Frequency	Applicability
a	Mass balance based on the excretion and the total (or total ammoniacal) nitrogen present at each manure management stage ⁽²⁾	Once every year ⁽³⁾ .	Generally applicable.
b	Emission factors ⁽²⁾	Once every year ⁽³⁾ .	Generally applicable.
⁽¹⁾ A description of the techniques is given in Section 4.8.2. ⁽²⁾ The monitoring technique is further developed at Member State level. ⁽³⁾ If feed intake, diet composition, animal performance and manure management practices are proven to be sufficiently stable, the monitoring frequency may be reduced to every time significant changes arise.			

In addition, the emerging technique in Section 5.4 has been identified.

3.7.3. Monitoring of methane emissions to air

UC 16. Methane emissions shall be monitored using the technique given below.

	Technique ⁽¹⁾	Frequency	Applicability
a	Emission factors ⁽²⁾	Once every year ⁽³⁾ .	Generally applicable.
⁽¹⁾ A description of the technique is given in Section 4.8.3. ⁽²⁾ The monitoring technique is further developed at Member State level.			

Technique ⁽¹⁾	Frequency	Applicability
⁽³⁾ If feed intake, diet composition, animal performance and manure management practices are proven to be sufficiently stable, the monitoring frequency may be reduced to every time significant changes arise.		

In addition, the emerging technique in Section 5.4 has been identified.

3.7.4. Monitoring of dust emissions to air

UC 17. Dust emissions from animal houses shall be monitored using the technique given below.

Technique ⁽¹⁾	Frequency	Applicability
a Emission factors ⁽²⁾	Once every year ⁽³⁾ .	Generally applicable.
⁽¹⁾ A description of the technique is given in Section 4.8.4. ⁽²⁾ The monitoring technique is further developed at Member State level. ⁽³⁾ If animal housing conditions are proven to be sufficiently stable, the monitoring frequency may be reduced to every time significant changes arise.		

Applicability

The applicability is restricted to cases where dust impact on sensitive receptors is expected and/or has been substantiated.

3.7.5. Odour monitoring

UC 18. Odour emissions shall be monitored using one of the techniques given below.

- (a) EN standards (e.g. dynamic olfactometry according to EN 13725 in order to determine the odour concentration and/or EN 16841-1 or -2 in order to determine the odour exposure);
- (b) Alternative methods (e.g. estimation of odour impact) for which no EN standards are available. In such a case, national or other international standards that ensure the provision of data of an equivalent scientific quality can be used.

Applicability

The applicability is restricted to cases where an odour nuisance at sensitive receptors is expected and/or has been substantiated.

3.7.6. Monitoring of the performance of air cleaning systems

UC 19. The performance of air cleaning systems shall be monitored using both of the techniques given below.

Technique ⁽¹⁾	Frequency	Applicability
a Verification of the air cleaning system performance by measuring ammonia, odour and/or dust under practical farm conditions and according to a prescribed measurement protocol and using EN	Once during the commissioning phase.	Not applicable if the air cleaning system has been verified in combination with a similar housing system and operating conditions.

Technique ⁽¹⁾		Frequency	Applicability
	standard methods or other national or international methods ensuring data of an equivalent scientific quality		
b	Control of the effective function of the air cleaning system (e.g. by continuously recording operational parameters or using alarm systems)	Monthly.	Generally applicable.
⁽¹⁾ A description of the techniques is given in Section 4.8.5.			

3.7.7. Monitoring of process parameters

UC 20. Process parameters shall be monitored using the techniques given below.

Technique ⁽¹⁾		Frequency	Applicability
a	Monitoring of water consumption at farm level and for the main water-consuming processes	Once every year ⁽²⁾ .	Monitoring of the main water-consuming processes separately may not be applicable depending on the configuration of the water supply network.
b	Monitoring of electric energy consumption		Monitoring the main energy-consuming processes separately may not be applicable, depending on the configuration of the energy supply network.
c	Monitoring of fuel consumption		Monitoring of fuel consumption in animal houses may not be applicable if a farm has a single heating plant that supplies heat to several animal houses as well as to residential buildings.
d	Monitoring of the number of animals including births and deaths		Generally applicable.
e	Monitoring of feed consumption		Generally applicable.
f	Monitoring of the quantity of manure generated		Generally applicable.
⁽¹⁾ A description of the techniques is given in Section 4.8.6.			
⁽²⁾ The monitoring is based on the records of process parameters kept for the whole period.			

4. DESCRIPTION OF TECHNIQUES

4.1. Nutritional management

	Technique	Description	Information on emissions
a	Limitation of the crude protein content in the diet by using a N-balanced diet based on the energy needs and digestible amino acids	The diet is balanced to meet the animal requirements of energy and digestible amino acids, avoiding excesses in crude protein supply, while taking into account their nutritional needs and welfare. Diets that are not properly balanced with the animal's requirements in terms of energy and digestible amino acids provide only the necessary quantities of the limiting nutrients, and overdoses of others. This results in an increase of excreted nutrients which results in higher emissions to the environment. Conversely, balanced diets provide the proper amount of amino acids, avoiding both excesses and deficiencies. This allows animals to realise their growth potential and to achieve optimal performance through maximised protein synthesis, while minimising emissions to the environment.	N.A.
b	Phase feeding with a diet formulation adapted to the specific requirements of the production period	The feed mix matches the animal requirements more accurately in terms of energy, amino acids, phosphorus and minerals, depending on the animal weight and/or production stage, while taking into account their nutritional needs and welfare. Feeding is typically divided into two to five phases, depending on the animal category, and can be done by groups of animals.	An ammonia reduction of at least 20 % for pigs and at least 10 % for poultry can be achieved in comparison to a reference system without the use of this technique.
c	Controlled supply of amino acids in crude-protein-reduced diets	A certain amount of protein-rich feedstuffs is substituted with low-protein feedstuffs, in order to further reduce the crude protein content, while taking into account their nutritional needs and welfare. The diet is supplemented with amino acids (e.g. lysine, methionine, threonine, tryptophan, valine) so that there is no deficiency in the amino acid profile.	N.A.

	Technique	Description	Information on emissions
d	Use of authorised feed additives which reduce the total nitrogen excreted / ammonia emissions	Authorised¹⁰ substances, microorganisms or preparations such as enzymes (e.g. NSP enzymes, proteases) or probiotics are added to feed or water in order to favourably affect feed efficiency, e.g. by improving the digestibility of feedstuffs or affecting the gastrointestinal flora. In addition to authorised substances that directly reduce the total nitrogen excreted, other additives may be used which have an indirect effect on ammonia emissions. For example, authorised¹¹ feed additives consisting of benzoic acid can be added to feed for fattening pigs (additives 4d210 and 4d211) and sows (additive 4d210) at rates between 0,5 % and 1 % complete feed with a moisture content of 12 %, in order to decrease urine pH.	In the case of benzoic acid addition, an ammonia reduction of at least 16 % can be achieved in comparison to a reference system without the use of this additive.
e	Use of authorised feed additives which reduce the total phosphorus excreted (e.g. phytase)	Authorised substances, microorganisms or preparations such as enzymes (e.g. phytase) are added to feed or water in order to favourably affect feed efficiency, e.g. by improving the digestibility of phytic phosphorus in the feedstuffs or affecting the gastrointestinal flora.	N.A.
f	Use of highly digestible inorganic phosphates to partially replace conventional sources of phosphorus in the feed	Highly digestible inorganic phosphates are used to replace conventional sources of phosphorus in the feed. These can be incorporated into animal feed (e.g. in powder or in granulated form) and are predictable in terms of their chemical composition and digestible phosphorus content.	N.A.

¹⁰ According to Regulation (EC) No 1831/2003 of the European Parliament and of the Council of 22 September 2003 on additives for use in animal nutrition (OJ L 268, 18.10.2003, pp. 29–43 ELI <http://data.europa.eu/eli/reg/2003/1831/oj>).

¹¹ According to the Commission Implementing Regulations (EU) 2016/900 and (EU) 2018/1550 of 8 June 2016 concerning the authorisation of benzoic acid as a feed additive for sows (holder of authorisation DSM Nutritional Product Sp. z o. o.) (OJ L 152, 9.6.2016, pp. 18–20 ELI http://data.europa.eu/eli/reg_impl/2016/900/oj) (additive 4d210) and 2024/1730 of 20 June 2024 concerning the authorisation of benzoic acid as a feed additive for weaned piglets and pigs for fattening (OJ L, 2024/1730, 21.6.2024, ELI: http://data.europa.eu/eli/reg_impl/2024/1730/oj) (additive 4d211)

	Technique	Description	Information on emissions
g	Precision feeding in pig rearing for individual animals	Precision feeding refers to continuous (e.g. daily) adjustment of feed nutrients, with levels that are closer to the real nutritional needs of each individual animal separately, maximising nutrient utilisation and thus reducing N and P excretion. Enteric methane emissions can also be targeted by this technique. The feed formulation can be obtained by mixing different feeds in the right proportions at the feeder level for each pig, which allows optimisation of performance and economics. This technique requires an appropriate individual animal identification system (e.g. chip), as well as specific measurements (e.g. weight). It also entails data processing and control actions at the individual feeder (e.g. a real-time algorithm to predict each animal's requirements and software to send instruction to the feeder). Furthermore, it requires sufficient space for an adequate feeder area.	N.A.

4.2. Housing systems

4.2.1. Types of floor and techniques to prevent or reduce ammonia and methane emissions in pig houses

Type of floor	Description
Fully slatted floor	A floor in the housing system where the whole pen area has metal, concrete or plastic-coated slats that allow faeces and urine to drop into a channel or a pit beneath.
Partly slatted floor	A floor in the housing system where both a slatted floor and a solid (concrete) floor are used in each pen or when there are different functional areas in the pen for reasons of animal welfare. Fouling of the solid floor is prevented by proper management of the indoor climate parameters, especially under hot conditions, and/or by proper design of the housing systems. For partly slatted floors covering 50 % of the area, an ammonia reduction of at least 15 % can be achieved in comparison to a “deep pit with fully slatted floor without any additional techniques” reference system.
Solid concrete floor	A floor in the housing system where the entire pen area consists of solid concrete. The floor can be covered with litter (e.g. straw) to varying degrees. The floor may be sloped to facilitate the drainage of urine.

	Technique	Description	Information on emissions
a	Deep pit (fully or partly slatted floor) combined with: (a) at least nutritional	Pens are equipped with a deep pit below the slatted floor that allows for the storage of the slurry	N.A.

	Technique	Description	Information on emissions
	management techniques in existing plants; and (b) additional mitigation measures (e.g. air cleaning system, slurry acidification, slurry cooling, use of slurry additives, frequent slurry removal) in new plants.	between infrequent removals. For fattening pigs, an overflow manure channel can be used. Removal of slurry for landspreading or to outdoor storage takes place as frequently as possible (e.g. at least every two months) unless there are technical restrictions (e.g. storage capacity). Deep pits are combined with at least nutritional management techniques (in existing plants) and with additional ammonia mitigation measures (e.g. air cleaning system, pH reduction of the slurry, slurry cooling) in new plants.	
b	Frequent slurry removal, e.g. by vacuum system, scraper, flushing (in the case of a fully or partly slatted floor)	Vacuum system: outlets at the bottom of the pit or channel are connected to a discharge pipe underneath which transfers slurry to outdoor storage. Slurry is frequently discharged by opening a valve or a plug in the main slurry pipe, e.g. once or twice every week; a slight vacuum develops and allows the complete emptying of the pit or channel. A certain depth of slurry needs to be obtained before the vacuum can be effective and the system can operate properly. A vacuum pump is not necessary if the slurry is flushed by gravity to a tank located on a level lower than the barn. From this tank the slurry can be pumped to a storage tank or to a mobile tanker. Scraper: there is a V-shaped channel with two inclined surfaces on each side of a central gutter, where urine can be drained to a collection pit through a drain in the bottom of the manure channel. From the pit, the solid fraction of the manure is extracted frequently (e.g. up to 12 times a day) by a scraper. The addition of a coating on the scraped floor is recommended in order to achieve a smooth(er) surface. Flushing: very frequent dilution and removal (e.g. once or twice per day) of the slurry is performed by flushing the channels with the liquid fraction of the slurry (dry matter content no higher than	Vacuum system: Ammonia reductions of at least 25 % (for removal at least twice per week) and 15 % (for removal at least every two weeks) can be achieved in comparison to a “deep pit with fully slatted floor without any additional techniques” reference system. Methane emissions are also reduced. Scraper: An ammonia reduction of at least 45 % can be achieved in comparison to a “deep pit with fully slatted floor without any additional techniques” reference system. Methane emissions are also reduced. Flushing: An ammonia reduction of at least 25 % can be achieved in comparison to a “deep pit with fully slatted floor without any additional techniques” reference system.

	Technique	Description	Information on emissions
		approximately 5 %). The liquid fraction of the slurry can also be aerated before flushing. Fresh water can also be used for flushing, although this results in higher water consumption. This technique can be combined with individual variations of the bottoms of channels or pits, e.g. gutters, tubes or a permanent slurry layer.	
c	Slanted walls or funnel system (in the case of a fully or partly slatted floor)	For slanted walls: the manure channel creates a V section with the point of discharge at the bottom. Manure removal is carried out at least twice every week. For funnel systems: plastic funnels are installed in rows beneath the slatted floor and they guide manure into a common manifold for rapid discharge by mechanically controlled flushing at pre-set intervals. The slope and the smoothness of the surface facilitate the slurry discharge.	An ammonia reduction of at least 40 % can be achieved in comparison to a “deep pit with fully slatted floor without any additional techniques” reference system. Methane emissions are also reduced.
d	Reduced manure pit (in the case of partly slatted floor)	The pen is equipped with a narrow pit with a width of about 0,6 m. The pit can be located in an external alley.	An ammonia reduction of at least 30 % can be achieved in comparison to a “deep pit with fully slatted floor without any additional techniques” reference system.
e	Full litter system (in the case of a solid concrete floor)	A fully concrete floor almost completely covered with a layer of straw or other lignocellulosic material. In the litter-floored system, solid manure is frequently removed (e.g. twice per week). Alternatively in the deep litter system, fresh straw is added on top and the accumulated manure is removed at the end of the rearing cycle. In order to include animal welfare considerations, separate functional areas can be organised into lying, feeding, walking and defecating areas.	N.A.
f	Kennel / hut housing (in the case of a partly slatted floor)	Separate functional areas are organised in the pens of naturally ventilated houses. The lying area (about 50 to 60 % of the total area) consists of a levelled insulated concrete floor with covered, insulated huts or kennels, with a hinged roof that can be raised or lowered to control temperature and ventilation. The activity and feeding areas have a slatted floor with a manure pit underneath and	An ammonia reduction of at least 33 % can be achieved in comparison to a “deep pit with fully slatted floor without any additional techniques” reference system.

	Technique	Description	Information on emissions
		frequent manure removal, e.g. by vacuum. Straw can be used on the solid concrete floor.	
g	Straw flow system (in the case of a solid concrete floor)	Pigs are reared in pens with solid floors, typically in a naturally ventilated house, where a sloped lying area and an excretion area are defined. Straw is provided to the animals daily. Pig activity pushes and distributes the litter down the pen's slope. The solid fraction can be removed frequently (e.g. daily) with a scraper.	An ammonia reduction of at least 33 % can be achieved in comparison to a “deep pit with fully slatted floor without any additional techniques” reference system. The technique may result in an increase in methane emissions.
h	Convex floor and separated manure and water channels (in the case of partly slatted pens)	Manure and water channels are built on opposite sides of the convex and smooth solid concrete floor. The water channel is installed underneath the side of the pen where the pigs tend to eat and drink. Water for cleaning the pens may be used to fill the water channels. The channel is partially filled with at least 10 cm of water. The manure channel can be built with flashed gutters or slanted walls which are normally flushed twice every day for example with water from the other channel or the liquid fraction of the slurry (dry matter content no higher than approximately 5 %).	An ammonia reduction of at least 40 % can be achieved in comparison to a “deep pit with fully slatted floor without any additional techniques” reference system.
i	Littered pens with combined manure generation (slurry and solid manure)	Farrowing pens are equipped with separate functional areas: a bedded lying area, walking and dung areas with slatted or perforated floors, and a feeding area on a solid floor. Piglets are provided with a littered and covered nest. Slurry is frequently removed with a scraper. Solid manure is manually removed from the solid floor areas on a daily basis. Litter is regularly provided. A yard can be combined with the system.	N.A.
j	Feeding/lying boxes on solid floor (in the case of litter-based pens)	Sows are kept in a pen divided into two functional areas, the main one littered and a series of feeding/lying boxes over a solid floor. Manure is captured in the straw or other lignocellulosic material, which is regularly supplied and replaced.	An ammonia reduction of at least 50 % can be achieved in comparison to a “deep pit with fully slatted floor without any additional techniques” reference system.
k	Manure pan (in the case of a fully or partly slatted floor)	A prefabricated pan (or pit) is positioned under the slatted floor. The pan is deepest at one end with a slope of at least 3 degrees towards a central manure channel; the manure discharges when its level reaches around 12 cm. If a	An ammonia reduction of at least 65 % can be achieved in comparison to a “deep pit with fully slatted floor without any additional techniques” reference system.

	Technique	Description	Information on emissions
		water channel exists, the pan can be subdivided into a water section and a manure section.	
l	Manure collection in water	Manure is collected in the cleaning water which is kept in the manure channel and refilled up to a level of around 120 to 150 mm. Slanted channel walls are optional. After each rearing cycle, the manure channel is emptied. To limit water consumption, other liquids (e.g. rinse water from room cleaning) may also be used instead of fresh water, provided that they have adequate physico-chemical and biological quality characteristics for this use.	An ammonia reduction of at least 20 % can be achieved in comparison to a “deep pit with fully slatted floor without any additional techniques” reference system.
m	V-shaped manure belts (in the case of a partly slatted floor)	V-shaped manure belts roll inside the manure channels covering the whole surface, so that all faeces and urine are dropped on them. Belts are run at least twice every day to separately carry urine and faeces to closed manure storage. Belts are made of plastic (polypropylene or polyethylene).	An ammonia reduction of at least 60 % can be achieved in comparison to a “deep pit with fully slatted floor without any additional techniques” reference system.
n	A combination of water and manure channels (in the case of a fully slatted floor)	The sow is kept in a fixed place (by using a farrowing crate) with a specific defecating area. The manure pit is split up into a wide water channel at the front and a small manure channel at the back, with a reduced manure surface. The front channel is partly filled with water.	An ammonia reduction of at least 50 % can be achieved in comparison to a “deep pit with fully slatted floor without any additional techniques” reference system.
o	Littered external alley (in the case of a solid concrete floor)	Animals are kept in a hybrid ventilated house, where a small door allows the pig to go out to defecate in an external alley with a concrete littered floor. The manure falls into a channel from where it is scraped once every day.	N.A.
p	Use of an air cleaning system, such as: (a) wet acid scrubber; (b) two-stage or three-stage air cleaning system; (c) bioscrubber (or biotrickling filter).	See Section 4.2.3.	
q	Use of floating balls in the manure channel	Balls half-filled with water and made of special plastic with non-sticky coating float on the surface of the manure channels.	An ammonia reduction of at least 25 % can be achieved in comparison to a “deep pit with fully slatted floor without any additional techniques” reference system. Methane emissions are also reduced.
r	Slurry cooling	A reduction of the slurry temperature (usually to reach 15°C) is achieved by installing a cooling system placed above the	For bottom cooling, an ammonia reduction of at least 20 % can be achieved in comparison to a “deep pit with fully slatted floor without

	Technique	Description	Information on emissions
		slurry (surface cooling with fins), above the concrete floor or cast into the floor (bottom cooling). The system consists of pipes in which a refrigerant or water is circulated. The pipes are connected to a heat exchange device to recover energy that may be used for heating other parts of the farm. The pit or the channels need to be emptied frequently due to the relatively small exchanging surface of the pipes. If waste heat or ground water cannot be used for cooling, the use of this technique may increase energy consumption.	any additional techniques” reference system. Methane emissions are also reduced. For fin cooling, an ammonia reduction of at least 45 % can be achieved in comparison to a “deep pit with fully slatted floor without any additional techniques” reference system. Methane emissions are also reduced.
s	Slurry acidification	Acid (e.g. sulphuric acid) is added to slurry in order to lower the pH to about 5,5 in the slurry pit. The addition can be carried out in a process tank, followed by aeration and homogenisation. Part of the treated slurry is pumped back to the storage pit under the housing floors. The treatment system is fully automated. Prior to (or after) landspreading on acid soils, lime addition may be required to neutralise the pH of the soil. Alternatively, acidification can be performed directly in the slurry store or continuously during landspreading.	An ammonia reduction of at least 60 % can be achieved in comparison to a “deep pit with fully slatted floor without any additional techniques” reference system. Methane emissions are also reduced.
t	Housing with distinct living zones for animals	Animals are raised in groups, in housing systems comprising at least three functional zones: lying/resting, feeding/activity and excretion. The surface area per pig is larger in comparison to conventional housing systems. An outdoor yard may also be available, where animals have access to fresh air and natural light, thus improving animal welfare. By having different functional zones, the pollutant characteristics of each zone can be targeted via specific techniques. For instance, the excretion zone can be in the outdoor yard with a small emitting surface and frequent manure removal (e.g. up to 12 times a day using a V-shaped scraper with a urine gutter / underfloor faeces-urine segregation). Normally, the animals keep the lying and feeding area clean (this natural behaviour is enhanced by natural/hybrid	An ammonia reduction of at least 75 % can be achieved in comparison to a “deep pit with fully slatted floor without any additional techniques” reference system.

	Technique	Description	Information on emissions
		ventilation for temperature differentiation between the excretion zone and the other zones). In comparison to conventional systems, the consumption of water per pig is typically higher (due to the greater surface area per pig), while the energy consumption is lower (due to natural/hybrid ventilation). The labour requirements are also typically higher (e.g. due to cleaning of straw from a larger area).	
u	Use of the following additives for the treatment of manure: (a) ion-exchange-based materials; (b) selected microorganisms.	Additives aimed at changing the physical, chemical and biological properties of slurry can be placed, manually or mechanically, on emitting (fouled) surfaces (e.g. slats, paved /solid floor, below slats), in the pits or in the manure storage facilities. The initial application is followed by periodic additions at varying frequency (determined by the active principle of each additive). Slurry additives include several categories, for example, complexes of bacteria or microorganisms, zeolites and chemical products. They act, for example, by changing the natural microbial balance, by inducing bio-acidification (production of organic acids) which lowers pH (thus reducing ammonia volatilisation), by limiting biological pathways leading to sulphide formation. This results in decreasing, for instance, ammonia, methane, hydrogen sulphide and/or odour emissions. In addition to the emission reduction, the additives can improve slurry handling by inducing liquefaction (lower viscosity) and/or crust weakening/breakdown (resulting in better pumpability and less foaming), thus reducing agitation needs, protecting equipment and limiting manual interventions (including those associated with hazardous H ₂ S exposure). - Ion-exchange-based materials (natural zeolites, activated zeolites or clay minerals) remove ammonia from slurry by trapping positive ions from solutions (NH ₄ ⁺) and replacing them with exchangeable	N.A.

	Technique	Description	Information on emissions
		positive ions (Na^+, Ca^{2+}) which are contained in the porous surfaces of the mineral crystalline structure. - Authorised microorganisms in the EFSA QPS (qualified presumption of safety) list (e.g. lactic acid bacteria) achieve conservation and stabilisation of livestock manure by spraying (liquid suspension) or dispersing (combination with drying powder). This can be done in the different stages of livestock rearing and production: on bedding/litter, in the pit (above/below), manure storage, on-farm processing of manure (e.g. composting, fermentation, acidification).	
v	Gas-permeable membranes (GPM) for ammonia capture from gaseous media	Microporous, hydrophobic membranes are used to capture ammonia from the air. The driving force for ammonia transfer across membranes is the concentration gradient. The process is carried out at atmospheric pressure. Once the ammonia is in the acidic solution (on the other side of the membrane), it combines with free protons to form non-volatile ammonium ions (NH_4^+). Various inorganic or organic acids can be used to prepare the acidic solution, such as sulphuric, hydrochloric, phosphoric and lactic acid. This way, ammonia contained in the air is transformed into a valuable ammonium fertiliser (e.g. ammonium sulphate, ammonium chloride, ammonium phosphate, ammonium lactate), which can also be transported off site. This technique is suitable for both natural and forced ventilation in the buildings, as it operates independently of the ventilation system. The system is usually located outside the housing and needs sufficient space for maintenance.	Ammonia reductions of at least 75 % (in pig farms) and 45 % (in poultry farms) can be achieved in comparison to a reference system without the use of this technique.

4.2.2. Housing systems and techniques to reduce ammonia emissions from poultry houses

Housing systems	Description
Cage systems	Housing systems for poultry built with sloping floors, made of welded wire mesh or plastic slats and equipped with fixtures and space for feeding, drinking, nesting, scratching, perching and egg collection. The cages are usually arranged on three or more tiers. Housing systems need to adhere to the requirements laid down in the latest version of Council Directive 1999/74/EC ¹² or the legislation replacing it.
Deep litter with manure pit	Housing systems for poultry with at least one third of the total floor area covered with litter (e.g. sand, wood shavings, straw). The remaining floor area is slatted, with a manure pit underneath. Feeding and drinking fixtures are located over the slatted area. Additional structures considered as outdoor areas, such as exercise yards, verandas, free-range areas or winter gardens can be present.
Aviaries	Housing systems for poultry divided into different functional areas for feeding, drinking, egg laying, scratching and resting. The usable area is increased by means of elevated slatted floors combined with tiers. The slatted area ranges between 30 % and 60 % of the total floor area. The remaining floor area is typically littered. Additional structures considered as outdoor areas, such as exercise yards, verandas, free-range areas or winter gardens can be present.

Techniques		Description	Information on emissions
All poultry species			
a	Use of an air cleaning system, such as: (a) wet acid scrubber; (b) two-stage or three-stage air cleaning system; (c) bioscrubber (or biotrickling filter).	See Section 4.2.3.	
b	Gas-permeable membranes (GPM) for ammonia capture from gaseous media	See Section 4.2.1.	
Laying hens, broiler breeders and pullets			
c	Manure removal by belts with at least: (a) one removal per week with air drying; or (b) two removals per week without air drying.	Belts are placed under the cages for manure removal. The frequency of removal can be once every week (with air drying) or more (without air drying). The collection belt may be ventilated for drying the manure. Whisk-forced air drying in the manure belt can be also used.	With two removals per week with air drying, an ammonia reduction of at least 30 % can be achieved in comparison to a “two removals per week without ventilation” reference system.
d	Forced or natural ventilation system without frequent manure removal in existing plants with deep litter and a manure pit,	The deep litter system (see above for description) is combined with infrequent manure removal, e.g. at the end of the rearing cycle. A	An ammonia reduction of at least 40 % can be achieved in comparison to a “deep litter / pit with partial litter” reference

¹²

Council Directive 1999/74/EC of 19 July 1999 laying down minimum standards for the protection of laying hens (OJ L 203, 3.8.1999, p. 53, ELI: <http://data.europa.eu/eli/dir/1999/74/oj>).

Techniques		Description	Information on emissions
	combined with at least an additional mitigation measure (e.g. achieving a high dry matter content of the manure, air cleaning system)	minimum dry matter content of manure of around 50 to 60 % is ensured. This is achieved by an appropriate ventilation system (e.g. fans and air extraction located at floor level).	system. The technique may result in an increase in methane emissions.
e	Manure belt or scraper (in the case of deep litter with a manure pit)	Manure is removed by scrapers (periodically) or by belts (once every week for dried manure, twice every week without drying).	For non-ventilated manure belts, an ammonia reduction of at least 70 % can be achieved in comparison to a “deep litter / pit with partial litter” reference system. For ventilated manure belts, an ammonia reduction of at least 80 % can be achieved in comparison to a “deep litter / pit with partial litter” reference system.
f	Forced air drying of manure via tubes (in the case of deep litter with a manure pit)	The deep litter system (see above for description) is combined with manure drying by means of forced ventilation applied through tubes that blow air (e.g. at 17 to 20 °C and 1,2 m ³ /bird) over the manure stored under the slatted floor.	An ammonia reduction of at least 45 % can be achieved in comparison to a “deep litter / pit with partial litter” reference system.
g	Forced air drying of manure using a perforated floor (in the case of deep litter with a manure pit)	The deep litter system (see above for description) is equipped with a perforated floor underneath the manure which allows for forced air blowing from below. The manure is removed at the end of the rearing cycle.	An ammonia reduction of at least 45 % can be achieved in comparison to a “deep litter / pit with partial litter” reference system.
h	Drying of litter on manure belts (in the case of aviaries)	Manure is collected on belts under the slatted floor and removed at least once every week by ventilated or not ventilated belts. Littered and solid floors can be combined in aviaries for pullets.	For non-ventilated manure belts, an ammonia reduction of at least 70 % can be achieved in comparison to a “deep litter / pit with partial litter” reference system. For ventilated manure belts, an ammonia reduction of at least 80 % can be achieved in comparison to a “deep litter / pit with partial litter” reference system.
i	Forced air drying of litter on manure belt (in the case of aviaries)	The technique reduces ammonia emissions from manure through targeted airflow around it, which slows down microbial activity (of breaking down uric acid in poultry manure) and limits the conversion of nitrogen compounds into volatile ammonia. Manure falls onto belts located beneath the slats, where it is dried using air (e.g. 0,3 m ³ of air per animal per hour at a minimum temperature of 20 °C) supplied through ducts under or beside the slatted floor. The airflow is strategically placed to ensure uniform drying,	An ammonia reduction of at least 62 % can be achieved in comparison to a “deep litter / pit with partial litter” reference system. Dust emissions are also reduced.

Techniques		Description	Information on emissions
		avoiding zones of high humidity. The system operates continuously, ensuring manure remains in a semi-dry state (e.g. dry matter content of at least 50 %) from the moment it is excreted until removal (at least once a week or twice per week if further drying is used).	
j	Forced drying of litter using indoor air (in the case of a solid floor with deep litter)	In a deep litter system without a manure pit, indoor air recirculation systems can be used to dry the litter, while meeting the physiological needs of the birds. To this end, fans, heat exchangers and/or heaters can be used.	An ammonia reduction of at least 25 % can be achieved in comparison to a “deep litter / pit with partial litter” reference system.
k	Litter slide (in the case of aviaries)	The technique keeps the thickness of the litter layer limited (e.g. maximum 3 cm) over the entire litter-covered floor surface, by frequently removing a portion of the litter layer with a slide.	An ammonia reduction of at least 15 % can be achieved in comparison to a “deep litter / pit with partial litter” reference system. Dust emissions are also reduced.
Broilers			
l	Forced drying of litter using indoor air (in the case of a solid floor with deep litter)	Indoor air recirculation systems can be used to dry the litter, while meeting the physiological needs of the birds.	An ammonia reduction of at least 25 % can be achieved in comparison to a “deep litter / pit with partial litter” reference system.
m	Forced drying of litter using heated indoor air (in the case of a solid floor with deep litter)	Ammonia emissions are reduced by drying the bedding with warm air and removing evaporated moisture with heated air (achieved by heat exchangers and/or heaters). Two configurations are possible: 1. Heating pipes are placed along the side wall inside the animal housing beneath the air inlet. The heated air from the pipes rises due to thermals and moves, together with the incoming air from the air inlets, towards the centre of the barn. 2. The warm air from the pipes, placed beneath air supply ducts in the roof (positive pressure ventilation), is mixed with the incoming air from the supply ducts and is evenly distributed over the entire surface of the barn which ensures uniform drying of the bedding. The heated air can absorb moisture evaporating from the bedding. A portion of the barn air is extracted so that the evaporated moisture is removed directly from the barn. The pipe heating system cannot be used in a covered run.	An ammonia reduction of at least 55 % can be achieved in comparison to a “deep litter / pit with partial litter” reference system.

Techniques		Description	Information on emissions
n	Natural or forced ventilation with a non-leaking drinking system (in the case of a solid floor with deep litter)	The solid floor is fully covered with litter which can be added to upon necessity. Floor insulation (using concrete, clay for example) prevents water condensation in the litter. Solid manure is removed at the end of the rearing cycle. The design and operation of the drinking water system prevents leakage and spillage of water on the litter. Additional structures considered as outdoor areas, such as exercise yards, verandas, free-range areas or winter gardens can be present. Natural ventilation can be combined with a free-range system.	N.A.
o	Heated and cooled littered floor (in the case of combideck systems)	A closed water circuit is installed below the floor and another is built at a deeper level for storing the excess heat or to return it to the poultry house when needed. A heat pump connects the two water circuits. At the beginning of the rearing period, the floor is heated with the stored heat in order to keep the litter dry by avoiding moisture condensation; during the second rearing cycle, birds produce an excess of heat which is preserved in the storing circuit while cooling down the floor, which reduces the breakdown of uric acid by reducing microbial activity.	An ammonia reduction of at least 40 % can be achieved in comparison to a “deep litter / pit with partial litter” reference system.
p	Litter composed of straw and sodium hydrogen sulphate	In order to reduce the pH value of litter, which in turn inhibits microbiological activity, a pellet made of a combination of 80 % straw (or other renewable material) and 20 % sodium hydrogen sulphate (NaHSO_4) is used as bedding.	An ammonia reduction of at least 58 % can be achieved in comparison to a “deep litter / pit with partial litter” reference system.
Ducks			
q	Frequent litter addition (in the case of a solid floor with deep litter or deep litter combined with a slatted floor)	Litter is kept dry by frequent addition (e.g. daily) of fresh material upon necessity. Solid manure is removed at the end of the rearing cycle. The housing system can be equipped with natural or forced ventilation and combined with a free-range system. In the case of deep litter combined with a slatted floor, the floor is equipped with slats in the drinker area (about 25 % of the total floor area).	N.A.
r	Frequent manure removal (in the	Slats cover the pit where the	Ammonia reductions of at least 15

Techniques		Description	Information on emissions
	case of a fully slatted floor)	manure is stored and evacuated to the external store. Frequent manure removal to an external store can be done: (a) by permanent gravity flow; (b) by scraping with variable frequencies. The housing system can be equipped with natural or forced ventilation and combined with a free-range system.	% (for permanent gravity flow) and 30 % (for scraping) can be achieved in comparison to a “deep litter / pit with partial litter” reference system.
Turkeys			
s	Natural or forced ventilation with a non-leaking drinking system (in the case of a solid floor with deep litter)	See description of technique (n) (in the broilers section).	
t	Forced drying of litter using heated indoor air (in the case of a solid floor with deep litter)	See description of technique (m) (in the broilers section).	

4.2.3. Treatment of emissions to air from animal housing

	Technique	Description	Information on emissions
a	Biofilter	The exhaust air is led through a filter bed of organic material, such as root wood or wood chips, coarse bark, compost or peat. The filter material is always kept moist by intermittent sprinkling of the surface. Odorous air compounds and dust particles and are absorbed by the wet film and are oxidised or degraded by microorganisms living on the moistened litter material. The inoculation of selected microorganisms mixtures may enhance the efficiency of biofilters. A sufficient area outside the animal house is needed to accommodate the filter packages.	An ammonia reduction of at least 70 % can be achieved in comparison to a reference system without the use of this technique. Methane emissions to air may also be reduced depending on operating conditions. N ₂ O emissions may potentially occur.
b	Bioscrubber (or biotrickling filter)	A packed tower filter with inert packing material which is normally maintained continuously wet by sprinkling water. Air pollutants are absorbed in the liquid phase and subsequently degraded by microorganisms settling on the filter elements. The inoculation of selected microorganisms mixtures may enhance the efficiency of biofilters. Partial air cleaning (e.g. to target a selected share of the exhaust air with a higher pollutant load) can also be applied in certain cases.	An ammonia reduction of at least 70 % can be achieved in comparison to a reference system without the use of this technique. Odour and dust emissions are also reduced.

c	Dry filter	The exhaust air is blown against a screen made of, for example, multi-layered plastic placed in front of the end wall ventilator. The passing air is subject to strong changes of direction, causing the separation of particles by centrifugal force, resulting in the reduction of dust emissions.	N.A.
d	Two-stage or three-stage air cleaning system	In a two-stage system, the first stage (wet acid scrubber) is usually combined with a bioscrubber (second stage). In a three-stage system, a first stage consisting of a water scrubber is usually combined with a second stage (wet acid scrubber), followed by a biofilter (third stage). Partial air cleaning (e.g. to target a selected share of the exhaust air with a higher load of pollutants) can also be applied in certain cases.	An ammonia reduction of at least 70 % can be achieved in comparison to a reference system without the use of this technique. Odour and dust emissions are also reduced.
e	Water scrubber	The exhaust air is blown through a packed filter medium by transverse flow. Water is continuously sprayed on the packing material. Dust is removed and settles in the water tank, which is emptied before refilling. This technique is also known as a “trickle bed reactor”.	N.A.
f	Water trap	The exhaust air is directed by ventilation fans down onto a water bath where dust particles get soaked. The flow is then redirected 180 degrees upward. The water level is topped up regularly to compensate for evaporation.	N.A.
g	Wet acid scrubber	The exhaust air is forced through a filter (e.g. packed wall) where a circulating acid liquid (e.g. sulphuric acid) is sprayed. Partial air cleaning (e.g. to target a selected share of the exhaust air with a higher load of pollutants) can also be applied in certain cases.	An ammonia reduction of at least 70 % can be achieved in comparison to a reference system without the use of this technique. Odour and dust emissions are also reduced.

4.2.4. Techniques to prevent or reduce dust and odour emissions and water and energy consumption in animal housing

Technique		Description
Dust emissions		
a	Prevention or reduction of dust generation inside livestock buildings by using, e.g., low-dust littering techniques, coarser litter material, litter slides, <i>ad libitum</i> feeding, low-dust feed, dust separators, optimised airspeed ventilation	Dust generation inside livestock building is prevented or reduced by applying techniques such as: (a) Application of fresh litter using a low-dust littering technique: littering techniques which release little dust are used, instead of techniques (e.g. blowers or mechanical distributors) that break up and disperse the litter; (b) Use of coarser litter material: coarser litter material, such as wood shavings, is used rather than finely structured material, such as chopped straw; (c) For litter slides, see description in Section 4.2.2; (d) <i>Ad libitum</i> feeding: free access to feed is provided, allowing the animal to self-regulate intake according to its biological needs; (e) Use of feed generating low dust: moist feed (e.g. soup or mash), pelleted feed or addition of oily raw materials or binders in dry feed systems are used, while ensuring a varied diet taking into account the proper

		functioning of the digestive system and the welfare of the animals; (f) Dust separators: dry feed stores which are filled pneumatically are equipped with dust separators, to capture dust that is generated during feed movement operations; (g) Optimised airspeed ventilation: the ventilation system is designed and operated to maintain a constantly optimised airspeed within the livestock housing environment. This prioritises consistent airflow patterns over short bursts of high-velocity ventilation (“purge cycles”), which may lead to rapid spikes in the airborne dust concentration.
b	Reduction of dust concentration inside livestock buildings by applying, e.g. water fogging, ionisation, oil spraying	The dust concentration inside livestock buildings is reduced by applying techniques such as: (a) Water fogging: water is sprayed by nozzles at high pressure to produce fine droplets that absorb heat and fall by gravity to the floor, moistening dust particles that become heavy enough to drop as well. Wet or moist litter needs to be avoided; (b) Ionisation: an electrostatic field is created in the house, resulting in an electrical charge (negative or positive) to dust. This causes the particles to adhere to oppositely charged surfaces or earthed surfaces, where they can be removed; (c) Oil spraying: pure vegetable oil is sprayed by nozzles inside the house. A mixture of water and around 3 % vegetable oil can also be used for spraying. Circulating dust particles are bound to the oil drops and collected in the litter. A thin layer of vegetable oil is also applied on the litter to prevent dust emissions. Wet or moist litter needs to be avoided.
c	Dust abatement using an air cleaning system, e.g. dry filter, two-stage or three-stage air cleaning system, water scrubber, water trap, wet acid scrubber, biofilter, bioscrubber	See Section 4.2.3.
Odour emissions		
d	Odour prevention and reduction measures inside the housing systems, e.g. keeping animals and surfaces dry and clean, reducing the manure-emitting surface, removing manure frequently, reducing the temperature of manure and the indoor environment, decreasing air flow	Odour emissions in the housing systems are prevented and/or reduced by applying techniques such as: (a) keeping the animals and the surfaces dry and clean (e.g. avoid feed spillages, avoid dung in lying areas with partly slatted floors); (b) reducing the emitting surface of manure (e.g. use metal or plastic slats, channels with a reduced exposed manure surface); (c) removing manure frequently to an external (covered) manure store (manure removal by flushing is avoided due to possible odour peaks); (d) reducing the temperature of the manure (e.g. by slurry cooling) and of the indoor environment; (e) decreasing the air flow and velocity over the manure surface; (f) keeping the litter dry and under aerobic conditions in litter-based systems.
e	Optimisation of discharge conditions of exhaust air from the animal houses, e.g. increasing the outlet height and/or ventilation velocity	Exhaust air is discharged from the animal houses by applying techniques such as: (a) increasing the outlet height (e.g. exhaust air above roof level, stacks, divert air exhaust through the ridge instead of through the low part of the walls); (b) increasing the vertical outlet ventilation velocity.

f	Odour abatement using an air cleaning system, e.g. bioscrubber (or biotrickling filter), biofilter, two-stage or three-stage air cleaning system)	See Section 4.2.3.
Water consumption		
g	Selection and use of suitable equipment (e.g. nipple drinkers, round drinkers, water troughs) for the specific animal category while ensuring water availability (<i>ad libitum</i>)	Water-efficient delivery systems (e.g. nipple drinkers, round drinkers, water troughs) are selected and used, taking into account the drinking behaviour of the different animal species. Optimal hydration and animal welfare is ensured through <i>ad libitum</i> access.
h	Verification and (if necessary) adjustment, on a regular basis, of the calibration of the drinking water equipment	The drinking water equipment is regularly checked and, if necessary, adjusted, to ensure consistent flow rates and functionality, while avoiding water spillages.
i	Use of techniques such as mechanical dry cleaning or high-pressure cleaners for animal housing and equipment	Animal housing and equipment are cleaned in a way that ensures hygiene while ensuring water efficiency, for example using pre-cleaning methods such as mechanical dry cleaning or using techniques such as high pressure-cleaners.
Energy consumption		
j	High-efficiency heating/cooling and ventilation systems	High-efficiency heating/cooling and ventilation systems employ technical solutions minimising energy consumption, while taking into account animal welfare requirements.
k	Optimisation of heating/cooling and ventilation systems and management, especially where air cleaning systems are used	Existing heating/cooling and ventilation systems are improved, while taking into account animal welfare requirements (e.g. concentration of air pollutants, appropriate temperatures). This can be achieved by measures such as: (a) implementation of automation and smart sensors; (b) replacement for high-efficiency motors; (c) use of frequency converters (e.g. for fans) and electronically commutated motors; (d) correct distribution of heating/cooling and ventilation equipment, temperature sensors and separate heated areas.
l	Insulation of the walls, floors and/or ceilings of animal housing	Insulation material can be naturally impermeable or provided with an impermeable coating. Permeable materials are provided with a vapour barrier installed, as humidity is a major cause of insulation material deterioration. A variant of insulation material for poultry farms can be heat-reflecting membranes, consisting of laminated plastic foils to seal off housing from air leakage and humidity.
m	Heat recovery (e.g. combideck system, heat recovery ventilation systems, slurry cooling heat recovery)	Recycling thermal energy from ventilation, waste products, or internal structures. A central unit uses an air-to-air heat exchanger to transfer heat from warm, stale exhaust air to cold, fresh incoming air. For the description of the combideck system, see Section 4.2.2.
n	Use of natural ventilation	Free (natural) ventilation in the animal house is generated by thermal effects and/or wind flow. The animal house can have openings in the ridge and, if necessary, also on the gable sides in addition to controllable openings in the side walls. The openings can be equipped with wind protection nets. Fan assistance can be used during hot weather.

o	Use of hybrid ventilation	Forced and natural ventilation are combined in different sections in the animal house. In the forced ventilated section, animals are protected from cold, heat, solar radiation, and rain. In the naturally ventilated section, animals have access to outdoor conditions (in the case of pigs, the excretion zone is typically located in the outdoor area).
p	Evaporative cooling systems using pads and water recirculation	Pad evaporative cooling systems use rigid cellulose, aspen or synthetic porous panels installed on one wall of the housing, continuously saturated with water by pumping. Water that is not evaporated or misted is collected and recirculated into the system. Tunnel fans draw outside air through the wet panels, evaporating water and cooling the air.

4.3. Manure storage

	Technique	Description	Information on emissions
a	Use of a storage facility with a sufficient capacity to hold the manure during periods in which landspreading is not possible	Storage facilities are designed with a sufficient capacity to hold the manure during periods in which landspreading is not possible. These periods depend on climatic conditions and the nitrogen requirements of the crops.	N.A.
b	Use of storage facilities that are able to withstand mechanical, chemical and thermal influences	Storage facilities are constructed using elements (e.g., steel with protective coatings, impermeable concrete, leakproof linings) which are able to withstand mechanical stress, chemical exposure, and thermal fluctuations, thus ensuring durability, environmental protection and operational reliability.	N.A.
c	Use of leakproof facilities and equipment for collection and transfer of slurry (e.g. pits, channels, drains, pump stations)	Facilities and equipment for slurry collection and transfer (e.g. pits, channels, drains, pump stations) are designed to prevent leaks and consequent contamination. This involves the use of suitable leakproof and corrosion-resistant materials.	N.A.
d	Verification of the structural integrity of storage facilities at least once every year	Regular checks of functional reliability (at least annual) are carried out and documented (via, for example, visual checks, photographs, test reports, leakage detection, observation of level change in storage filled with water, data obtained by sensors), including verification of the tightness of all components of storage facilities, pipes and fittings, to ensure leakproof integrity and operational standards.	N.A.
e	Coverage ensuring the highest environmental performance, with one of the	Stores are covered with rigid, flexible or floating covers. Rigid covers can be made of concrete, fibreglass panels or polyester sheets with a flat deck or conical shape, applied to	Rigid covers: An ammonia reduction close to 100 % can be achieved in comparison to a store with no cover.

	Technique	Description	Information on emissions
	following techniques: (a) rigid covers; (b) flexible covers; (c) floating covers.	concrete or steel tanks and silos. The covers are well-sealed and 'tight' to minimise air exchange and to prevent rain and snow from entering. Flexible covers include: (a) tent covers, equipped with a central supporting pole and spokes radiating from the tip; a fabric membrane is spread over the spokes and tied to a rim brace; openings are kept to a minimum; (b) dome-shaped covers, with a curved structural frame installed over round stores with the use of steel components and bolted joints; (c) flat covers, consisting of a flexible and self-supporting composite material held by plugs on a metal structure. Floating covers include: (a) crust layers: natural or straw-induced crusts (at least 2 % dry matter content) form on slurry surfaces; they require undisturbed, thick layers and filling from below the surface once the cover is formed; (b) floating bulk materials or modular systems: these include: polystyrene pellets, light expanded clay aggregates, perlite, zeolite, floating hexagonal plastic tiles; agitation of the slurry during stirring, filling and emptying may preclude the use of some floating materials which may cause sedimentation or blockages in the pumps; the choice of the materials needs to take into account possible manure contamination; (c) air-inflated covers: a cover made of PVC fabric supported by an inflatable pocket floats over the slurry; (d) floating flexible covers: plastic floating covers (e.g. blankets, canvas, films) rest on the slurry surface; (e) flexible plastic sheet: impermeable plastic sheet (e.g. HDPE) used for earth-banked storage, secured at the bank tops and supported on floats; it can be fitted with collection piping for removal of gases and systems for rainwater collection.	Tent covers: An ammonia reduction of at least 85 % can be achieved in comparison to a store with no cover. Dome-shaped covers: An ammonia reduction of at least 80 % can be achieved in comparison to a store with no cover. Flat covers: An ammonia reduction of at least 60 % can be achieved in comparison to a store with no cover. Crust layers: An ammonia reduction of at least 20 % can be achieved in comparison to a store with no cover. Floating bulk materials: An ammonia reduction of at least 40 % can be achieved in comparison to a store with no cover. Air-inflated covers: An ammonia reduction of at least 60 % can be achieved in comparison to a store with no cover. Floating flexible covers: An ammonia reduction of at least 80 % can be achieved in comparison to a store with no cover. Flexible plastic sheet: An ammonia reduction of at least 60 % can be achieved in comparison to a store with no cover.
f	Installation of a leakage detection system	A system using, for example, a geomembrane, an underlying drainage layer, and a drainage pipe system enabling early detection (e.g. with a piezometer) and	N.A.

	Technique	Description	Information on emissions
		containment of breaches.	
g	Minimisation of stirring of slurry	The stirring of slurry is kept to a minimum. This involves: (a) filling the store below surface level; (b) discharging as close as possible to the base of the store; (c) avoiding unnecessary homogenisation and circulation of slurry (before emptying the slurry store).	N.A.
h	Appropriate design of the slurry store by using techniques such as: (a) reduction of the ratio between the emitting surface area and the volume of the slurry store; (b) reduction of wind velocity and air exchange on the slurry surface by operating the store at a lower level of fill	Storage facilities are designed taking into account structural and operational features aimed at reducing emissions to air. This includes reducing the ratio of emitting surface area to slurry volume, managing fill levels to limit wind velocity and air exchange over the slurry surface, and increasing freeboard to create a windshield effect that suppresses turbulent air movements.	N.A.
i	Slurry acidification	See Section 4.2.1.	
j	Slurry cooling	See Section 4.2.1.	
k	Impermeable base and walls, e.g. with clay or plastic lining (or double-lined), and leakproof structures	Earth-banked storage is constructed with leakproof structures (e.g. joints, fixtures) and with impermeable bases and walls using materials like compacted clay, high-density polyethylene (HDPE) plastic lining, or double-lined systems (e.g. geomembrane, clay) to prevent leakage and contamination of groundwater.	N.A.
l	Storage of dried solid manure in a barn	The barn is usually a simple construction with an impermeable floor and a roof, with sufficient ventilation to avoid anaerobic conditions and an access door for vehicles. Dried poultry manure (e.g. litter from broilers and laying hens, air-dried laying hen excreta collected on belts) can typically	N.A.

	Technique	Description	Information on emissions
		be transported by belts or front-end loaders from the poultry house to the barn where it can be stored for a long period of time without the risk of remoistening.	
m	Use of a concrete silo for storage of solid manure	A foundation slab of water-impermeable concrete that can be combined with walls on three sides and with a cover (e.g. roofing over the manure platform, UV-stabilised plastic). The floor is sloped (e.g. 2 %) towards a front drain gutter. Liquid fractions and any run-off caused by rainfall are collected in a leak-tight concrete pit and handled afterwards.	N.A.
n	Storage of solid manure on a solid impermeable floor designed to prevent leaks to the environment	The storage is equipped with a solid impermeable floor, a drainage system, and connected to a tank for collection of liquid fractions and any run-off caused by rainfall.	N.A.
o	Appropriate distances between temporary solid manure heaps and water bodies, with annual change of location and minimisation of time on field	When solid manure is stacked directly on the soil in the field prior to landspreading, the time of manure on field is minimised and the storage location is changed at least every year and situated as far as possible from water bodies.	N.A.
p	Reduction of the ratio between the emitting surface area and the volume of the solid manure heap	The ratio between the emitting surface area and the volume of the heap can be reduced, for example through manure compaction or using a three-sided wall store.	N.A.
q	Coverage of temporary solid manure heaps	Solid manure heaps are covered with materials such as UV-stabilised plastic covers, peat, sawdust, or wood chips. Tight covers decrease air exchange and aerobic decomposition in the manure heap.	N.A.

4.4. On-farm processing of manure

	Technique	Description
a	Mechanical separation of slurry, e.g. using: (a) screw press separator; (b) decanter-centrifuge separator; (c) separation by sieves; (d) filter pressing.	Separation of liquid and solid fractions with different dry matter content, using for example screw press separators, decanter-centrifuge separators, separation by sieves or filter pressing. Separation can be enhanced by additives or by applying different temperatures/pressures. The technique is applied, for example, when a reduction of the nitrogen and phosphorus content in the solid phase is needed due to limited available land for manure application.

b	Anaerobic digestion	Anaerobic microorganisms decompose the organic matter of manure in a closed reactor in the absence of oxygen. Biogas is produced and collected to serve energy generation, i.e. production of heat, combined heat and power, and/or transport fuel. Some of the heat produced is recycled in the process. The stabilised residue (digestate) can be used as fertiliser (with sufficiently solid digestate after composting). Solid manure can be co-digested with slurry and/or other co-substrates, while ensuring a dry matter content lower than 18 %. The inoculation of selected microorganisms mixtures may enhance the speed and efficiency of the process.
c	Manure drying with an external tunnel	Manure is collected from the laying hen houses and removed by belts that convey it outdoors to a dedicated closed structure, containing a series of perforated overlapping belts that form the tunnel. Warm air is blown through the belts, drying the manure in about two or three days. The tunnel is ventilated with air extracted from the laying hens' house.
d	Mechanical separation of slurry into solid and liquid fractions, followed by aerobic digestion (aeration) of the liquid fraction, with nutrient recovery	The slurry is separated into liquid and solid fractions using mechanical separation equipment and nutrients are recovered when the solid and liquid fractions are used for crop nutrition. Most of the phosphorus and a substantial portion of the nitrogen are retained in the solid fraction, which is a product with lower volume and weight. The liquid fraction is aerated, intermittently and at low intensity, by means of submerged or floating aerators in a continuous or batch process which induces biological oxidation of volatile organic compounds. The process is managed through the control of operating parameters, such as aeration intensity (dissolved oxygen concentration), pH increase limitation, temperature (e.g. between 45 °C and 75 °C) and minimising slurry mixing, with the objective of decreasing ammonia (NH₃) losses to the atmosphere, while suppressing anaerobic conditions and methane formation. Ammonia released during aeration can be optionally collected (see for example the technique “Gas-permeable membranes (GPM) for ammonia capture from gaseous media”). Foam and sludge, which contain some nutrients (e.g. P) captured from the liquid fraction, are collected from the reactor and dried using mechanical separators, and then managed with the rest of the solid fraction. The technique is typically applied where pathogen and odour reduction is important prior to landspreading. Care should be taken to address possible cross-media effects.
e	Mechanical separation of slurry into solid and liquid fractions, followed by nitrification-denitrification of the liquid fraction, with nutrient recovery	The slurry is separated into liquid and solid fractions using mechanical separation equipment and nutrients are recovered when the solid and liquid fraction are used for crop nutrition. Most of the phosphorus and a substantial portion of the nitrogen are retained in the solid fraction, which is a product with lower volume and weight. The liquid fraction is subjected to nitrification via biological treatment under controlled aerobic conditions, in which ammonia (NH₃) is oxidised to nitrites (NO₂⁻) and nitrates (NO₃⁻), thus reducing ammonia losses to the atmosphere. Coagulants and flocculants are added to enhance the process. The effluent, rich in nitrates, and also containing any remaining P, as well as K, Na and Ca, may be reused in agriculture (e.g. through fertigation). Alternatively, it can be forwarded to the denitrification stage, in which nitrite and nitrate are transformed into N₂ in the presence of organic carbon. Some of the microorganisms can be recirculated to the aeration basin to maintain biological activity. Foam and sludge, which contain some nutrients (e.g. P) captured from the liquid fraction, are collected from the basin(s) and dried using mechanical separators, and then managed with the rest of the solid fraction. The nitrification-denitrification process can be achieved in a single basin or multiple basins in series. Through the setting of the retention time in the basin(s), the degree of nitrification-denitrification, including the concentration of the nitrates in the effluent, can be controlled. Care should be taken to address possible cross-media effects.

f	Composting of solid manure	Solid manure, including the solid fraction of the slurry, is decomposed by microorganisms under aerobic conditions, producing a final product (compost) sufficiently stable for transport, storage and landspreading. Biological inoculum, straw, sawdust, green residues or other organic wastes (e.g. digestate) can be co-composted with solid manure. Odour, microbial pathogens and water content of manure are reduced. The supply of oxygen is achieved by mechanical reversal of the windrows or by forced aeration of the heaps. Drums and composting (vertical and horizontal) tanks/silos can also be used. The inoculation of selected microorganism mixtures may enhance the speed and efficiency of the process. The technique can be used, for example, when pathogen and odour reduction is important prior to landspreading. Windrows need sufficient space to be established on the farm. Other composting options (e.g. tanks/silos, drums) usually require less space.
g	Vacuum evaporation with ammonia recovery	Evaporation is used to reduce digestate volume (by up to 70 %), resulting in concentrated nutrients. The liquid phase of the manure is typically evaporated under vacuum conditions at low temperatures (e.g. approximately 60 °C). During the evaporation process, gaseous ammonia is bound as an ammonium sulphate solution. Vacuum evaporation is primarily a method for reducing the volume of farm manure.
h	Vacuum fine separators with screens	The technique consists of a continuous fine separation technology designed for nutrient-optimised processing of liquid nutrient-rich suspensions (e.g. slurries, digestates). It enables solid-liquid separation into an energy-rich solid fraction and a nutrient-reduced liquid phase, suitable for further membrane filtration or fertiliser processing. This technique is suitable for liquids with low dry matter content (e.g. 1 %) using, for instance, fibre-retaining screens with different aperture sizes. It can be used in combination with pre-separation techniques.
i	Pressure and flocculation solid-liquid separators	A slurry treatment technique consisting of several stages. First, manure is divided into a solid and a liquid fraction. The liquid fraction is then subjected to a physico-chemical treatment complemented by decantation, which results in clarification and removal of suspended solids. Afterwards, the treated liquid is acidified to reduce ammonia emissions. In parallel, the solid fraction and sludge generated throughout the different stages of the liquid treatment undergo a dewatering process and are dried using solar energy. An ammonia reduction of at least 63 % can be achieved in comparison to a reference system without the use of this technique.
j	Mechanical separation of slurry in a solid and liquid fraction, followed by treatment in constructed wetlands of the liquid fraction, with nutrient recovery.	The slurry is separated into liquid and solid fractions using mechanical separation equipment. If there is enough space, the separation may be further enhanced by a subsequent decantation step. Most of the phosphorus and a substantial portion of the nitrogen are retained in the solid fraction, which is a product with lower volume and weight. The liquid fraction flows to the constructed wetland, where it percolates the subsurface through the gravel bed planted with specialised plant species (e.g. <i>Phragmites australis</i>, <i>Juncus</i>, <i>Typha</i>, <i>Apium nodiflorum</i>), without flooding the sand surface. Nitrogen (as NO_3^- and NH_4^+) is absorbed by the plants, thus reducing ammonia (NH_3) losses to the atmosphere. Possible emissions of nitrous oxide (N_2O) are kept low with proper management of the operational parameters. Suspended solids and Biological Oxygen Demand (BOD) are also reduced. The treated effluent exits the constructed wetland by gravity into a collection pond through a drainage pipe and can be reused in agriculture (e.g. through fertigation), since it contains any remaining N and/or P. No energy consumption is needed for the operation of the constructed wetlands. The system is adequately sized (hydraulic retention time is 3-5 days) to handle the whole volume of the liquid fraction and is designed to prevent any leakage to the environment (i.e. soil and groundwater), by incorporating double-layer plastic impermeabilisation. Periodic harvesting of vegetation/biomass, which can be used as co-substrate for anaerobic

		digestion or as feedstock for composting, and removal of the accumulated organic matter are done as part of proper management. Care should be taken to address possible cross-media effects.
--	--	--

4.5. On-farm manure landspreading

	Technique	Description	Information on emissions
a	Sufficient distance between landspreading areas and areas at risk of leaching/run-off, as well as neighbouring areas	Sufficient distance is left between landspreading areas and: i) areas where there is a risk of leaching and/or run-off to water (e.g. watercourses, springs, boreholes) and ii) neighbouring properties (including hedges).	N.A.
b	Application restrictions during run-off risk conditions	Manure landspreading is avoided when the risk of leaching and/or run-off is significant. This takes into account e.g. soil type and conditions, slope of the field, climatic conditions, field drainage and irrigation, crop type and conditions, water resources. In particular, manure is not applied when: i) the field is flooded, frozen or snow-covered; ii) the risk of run-off is high due to specific soil conditions (e.g. water saturation or compaction) and with the slope of the field.	N.A.
c	Landspreading taking into account crop nutrient requirements	Manure landspreading is carried out taking into account the nitrogen and phosphorus requirements, manure nutrient content, and soil characteristics, thus preventing and/or reducing nutrient losses into the environment.	N.A.
d	Slurry acidification prior to application	Slurry is subject to acidification prior to landspreading with a suitable spreader (e.g. band spreader, shallow injector, deep injector). A description of slurry acidification is given in Section 4.2.1.	N.A.

	Technique	Description	Information on emissions
e	Use of diluted slurry in low-pressure water irrigation systems	Slurry is diluted (to < 2 % dry matter content) in water, waste water (see Section 4.7.3) if the physico-chemical and biological characteristics of the waste water are adequate for such use (e.g. the water is not contaminated by detergents) or, where authorised, reclaimed water compliant with Regulation (EU) 2020/741, provided that the intended use is addressed in the water reuse risk management plan and complies with applicable permit or national authorisation conditions, to achieve faster infiltration in the soil and thus reduce ammonia emissions (see Section 4.7.3). The dilution rate (water:slurry) is typically in the range of 1:1 to 50:1. Diluted slurry can be injected into irrigation pipelines and pumped under low pressure to the irrigation system (e.g. sprinkler or travelling irrigator).	An ammonia reduction of at least 30 % can be achieved in comparison to spreading over the whole soil surface ("broadcast") without incorporation.
f	Slurry application through band spreaders, using one of the following techniques: (a) trailing hose; (b) trailing shoe.	Band spreading is a precision application method delivering slurry in narrow bands rather than uniform broadcasting. A trailing hose consists of a series of flexible hoses hanging from a wide bar mounted onto the slurry trailer. The hoses discharge slurry at ground level in wide parallel bands, enabling application between the rows of arable crops. A trailing shoe consists of a series of rigid pipes which terminate in metal 'shoes', designed to apply slurry directly in narrow bands to the soil surface and below the crop canopy. Some types of trailing shoes are designed to cut a shallow slit in the soil to aid infiltration.	Trailing hose: An ammonia reduction of at least 30 % can be achieved in comparison to spreading over the whole soil surface ("broadcast") without incorporation. Trailing shoe: An ammonia reduction of at least 40 % can be achieved in comparison to spreading over the whole soil surface ("broadcast") without incorporation.
g	Slurry application through shallow injectors (open slot)	An injector using tines, knives or discs to cut vertical slots (typically 4 to 6 cm deep) in the soil, forming grooves into which slurry is deposited. The injected slurry is fully or partially placed below the soil surface and grooves will normally be left open after slurry application.	An ammonia reduction of at least 60 % can be achieved in comparison to spreading over the whole soil surface ("broadcast") without incorporation.
h	Slurry application through deep injectors (closed slot)	An injector using tines, knives or discs to cultivate the soil and deposit slurry into it, before covering the slurry fully by means of press wheels or rollers. The depth of the closed slot ranges between 10 cm and 20 cm.	Ammonia reductions of at least 80 % (for slots < 15 cm) and 90 % (for slots ≥ 15 cm) can be achieved in comparison to spreading over the whole soil surface ("broadcast") without incorporation.

	Technique	Description	Information on emissions
i	Manure incorporation in the soil as soon as possible after application (with a time delay up to four hours or up to 12 hours when conditions are not favourable for a faster incorporation)	Manure incorporation involves mixing or covering surface-applied manure into the soil to minimise nutrient losses. This is done by different tillage practices (e.g. cultivators, disc harrows) depending on the soil type and conditions. Manure is completely mixed with soil or buried. Incorporation is either immediate or with a maximum time delay up to four hours. The time delay can be up to 12 hours when conditions are not favourable for a faster incorporation (i.e. when human and machinery resources are not available and/or in case of adverse climate conditions). Incorporation takes place after spreading via a suitable spreader (e.g. rota-spreader, rear discharge spreader, dual-purpose spreader for solid manure, band spreader for slurry).	Ammonia reductions of at least 90 % (for immediate slurry incorporation by primary tillage practices), 70 % (for immediate slurry incorporation by non-inversion cultivation), 65 % (for immediate solid manure incorporation by non-inversion cultivation), 55 % (for slurry incorporation in less than 4 hours), 50 % (for solid manure incorporation in less than 4 hours), 45 % (for slurry incorporation between 4 and 12 hours), 35 % (for solid manure incorporation between 4 and 12 hours), 30 % (for slurry incorporation beyond 12 hours) and 25 % (for solid manure incorporation beyond 12 hours) can be achieved in comparison to spreading over the whole soil surface ("broadcast") without incorporation.

4.6. On-farm storage of dead animals

Technique	Description
Use of authorised storage systems	Fallen stock that is not immediately collected by an authorised collector, or immediately treated, is stored in authorised storage systems, such as closed, leakproof containers. Refrigeration may be necessary, depending on temperature and on frequency of removal (e.g. fallen stock is collected less frequently than once every week). The storage systems are located in line with relevant legislation.

4.7. Farm-wide activities

4.7.1. Odour emissions

Technique	Description
Appropriate distances between the farm/plant and sensitive receptors	At the planning stage of the plant/farm, appropriate distances between the plant/farm and sensitive receptors are ensured by applying minimum standard distances or performing dispersion modelling to predict/simulate odour concentration in surrounding areas.
Prevention and reduction of odour emissions in animal housing	See Section 4.2.2.
Prevention and reduction of odour emissions from manure storage	See Section 4.3.
Prevention and reduction of odour emissions from manure processing	See Section 4.4.

Technique	Description
Prevention and reduction of odour emissions from manure landspreading	See Section 4.5.

4.7.2. Noise emissions

	Technique	Description
a	Appropriate distances between the plant/farm and sensitive receptors	At the planning stage of the plant/farm, appropriate distances between the plant/farm and sensitive receptors are ensured by applying minimum standard distances.
b	Appropriate equipment location	This includes techniques such as: (a) increasing the distance between the emitter and the receiver; (b) minimising the length of feed delivery pipes; (c) locating feed bins and feed silos so as to minimise the movement of vehicles on the farm.
c	Operational measures	These include measures such as: (a) closure of doors and major openings of the building, especially during feeding time, if possible; (b) equipment operation by experienced staff; (c) avoidance of noisy activities at night and during weekends, if possible; (d) provisions for noise control during maintenance activities; (e) operation of conveyers and augers full of feed, if possible; (f) keeping outdoor scraped areas to a minimum in order to reduce noise from scraper tractors.
d	Low-noise equipment	This includes equipment such as: (a) high-efficiency fans, when natural ventilation is not possible or sufficient; (b) pumps and compressors; (c) feeding system which reduces the pre-feeding stimulus (e.g. holding hoppers, passive <i>ad libitum</i> feeders, compact feeders).
e	Noise-control equipment	This includes: (a) noise reducers; (b) vibration isolation; (c) enclosure of noisy equipment (e.g. mills, pneumatic conveyers); (d) soundproofing of buildings.
f	Noise abatement techniques	Inserting obstacles between emitters and receivers.

4.7.3. Water management

	Technique	Description
a	Detection and repair of water leaks	Regular visual checks, pressure tests, or moisture sensors can be used to identify and promptly repair water leaks.
b	Drainage of waste water to a dedicated container or to a slurry store	Waste water is directed to sealed, leakproof containers to hold it. Waste water can also be directed to slurry stores provided that it is of sufficient quality not to interfere with the physico-chemical and biological characteristics of the slurry.
c	Segregation of uncontaminated rainwater from waste water streams that require treatment	Segregation is carried out by implementing separate collection in the form of properly designed and maintained drainage systems.
d	Reuse of uncontaminated rainwater as cleaning water	Collected rainwater that has not entered into contact with manure and/or hazardous chemical substances may be utilised for cleaning, provided it meets hygiene requirements.

e	Keeping the fouled yard areas as small as possible	Keeping the fouled yard area as small as possible minimises the volume of water needed for cleaning and the resulting volume of waste water.
f	Treatment of waste water	Treatment can be performed by sedimentation and/or biological treatment. For waste water with a low pollutant load, treatment can be carried out by means of swales, ponds, constructed wetlands, soakaways, etc. A first flush system can be used for separation before biological treatment.
g	Use of waste water in irrigation and landspreading	Waste water streams can be settled, e.g. in tanks or earth-banked storage, before landspreading. The resulting solid fraction can also be landspread. Waste water can also be mixed with slurry in order to obtain diluted slurry to be used for irrigation and landspreading. The liquid can be pumped from the stores and brought into a pipeline that goes to a sprinkler or travelling irrigator for example, which landspreads the water at a low application rate. Irrigation can also be carried out using equipment with controlled application to ensure a low trajectory (low spread pattern) and large droplets. Waste water needs to have adequate physico-chemical and biological quality characteristics for this use.

4.7.4. Energy management

	Technique	Description
a	Use of energy-efficient lighting	More energy-efficient lighting can be attained by the following: (a) Replacing conventional tungsten light bulbs or other low-efficiency light bulbs with more energy-efficient lights such as fluorescent, sodium, and LED lights; (b) Using devices to adjust the frequency of micro flashes, dimmers to adjust artificial lighting, sensors or room entry switches to control the lighting; (c) Daylight optimisation by allowing more natural light to enter, e.g. by using vents or roof windows. Natural light has to be balanced with potential heat losses; (d) Applying lighting schemes, using a variable lighting period.
b	Use of heat exchangers. One of the following systems may be used: - air-air; - air-water; - air-ground	In an air-air heat exchanger, the incoming air absorbs heat from the exhaust air from the plant. It can be composed of plates of anodised aluminium or PVC tubes. In the air-water heat exchanger, water flows through aluminium fins located in the exhaust ducts and absorbs heat from the exhaust air. In the air-ground heat exchanger, fresh air is circulated through buried pipes (e.g. at a depth of about 2 metres) taking advantage of the low seasonal temperature variation of soil.
c	Use of heat pumps for heat recovery	Heat is absorbed from various media (water, slurry, ground, air, etc.) and transferred to another location, via a fluid circulated in a sealed circuit using the reverse refrigeration cycle principle. The heat can be used to produce sanitised water or to feed a heating system or a cooling system. The technique can absorb heat from various circuits, such as slurry cooling systems, geothermal energy, scrubbing water, slurry biological treatment reactors, or biogas engine exhaust gases.
d	Renewable energy generation	The installation of site-appropriate renewable energy generators on buildings and land, including their corresponding battery storage where appropriate.
e	Insulation of the walls, floors and/or ceilings of buildings/structures	Insulation material can be naturally impermeable or provided with an impermeable coating. Permeable materials are provided with a vapour barrier installed, as humidity is a major cause of insulation material deterioration.

4.8. Monitoring

4.8.1. Monitoring of nitrogen and phosphorus excretion

	Technique	Description
a	Mass balance of nitrogen and phosphorus based on the feed intake, crude protein and total phosphorus content in the diet, and animal performance	The mass balance is calculated for each animal category reared on the farm, coinciding with the end of a rearing cycle, on the basis of the following equations: $N_{\text{excreted}} = N_{\text{diet}} - N_{\text{retention}}$ $P_{\text{excreted}} = P_{\text{diet}} - P_{\text{retention}}$ N_{diet} is based on the feed input and on the crude protein content of the diet. P_{diet} is based on the feed input and on the total phosphorus content of the diet. The crude protein and the total phosphorus contents can be obtained by one of the following methods: (a) in the case of external feed supply: in the accompanying documentation; (b) in the case of self-processing of feed: by sampling of feedstuff compounds from the silos or the feeding system to analyse the total content of phosphorus and crude protein or, alternatively, in the accompanying documentation or using standard values for the total content of phosphorus and crude protein of the feedstuff compounds. $N_{\text{retention}}$ and $P_{\text{retention}}$ can be estimated by one of the following methods: (a) statistically derived equations or models; (b) standard retention factors for the nitrogen and phosphorus contents of the animal (or of eggs, in the case of laying hens); (c) analysis of the nitrogen and phosphorus contents of a representative sample of the animal (or of eggs, in the case of laying hens). The mass balance considers in particular any significant changes to the diet commonly applied (e.g. change of a compound feed). The monitoring technique is further developed at Member State level. In order to apply the monitoring technique, farms need to keep record of information on key parameters (according to the methodology defined by the Member State), e.g. numbers of animals, feed intake, crude protein and total phosphorus content in the diet, quantity of manure generated.

4.8.2. Monitoring of ammonia emissions to air

	Technique	Description
a	Mass balance based on the excretion and the total (or total ammoniacal) nitrogen present at each manure management stage	Ammonia emissions are estimated based on the amount of nitrogen excreted by each animal category and using the total nitrogen (or the total ammoniacal nitrogen – TAN) flow and the volatilisation coefficients (VC) over each manure management stage (housing, storage, landspreading). The equations applied for each of the manure management stages are: $E_{\text{housing}} = N_{\text{excreted}} \cdot VC_{\text{housing}}$ $E_{\text{storage}} = N_{\text{storage}} \cdot VC_{\text{storage}}$ $E_{\text{spreading}} = N_{\text{spreading}} \cdot VC_{\text{spreading}}$ $E_{\text{processing}} = N_{\text{processing}} \cdot VC_{\text{processing}}$ where: (a) E is the annual NH_3 emission from the animal house, manure storage or landspreading (e.g. in $\text{kg NH}_3/\text{animal place/year}$); (b) N is the annual total nitrogen or TAN excreted, stored or applied in landspreading (e.g. in $\text{kg N}/\text{animal place/year}$). If appropriate, nitrogen

	Technique	Description
		additions (e.g. related to litter, recycling of scrubbing liquids) and/or nitrogen losses (e.g. related to manure processing) can be considered; (c) VC is the volatilisation coefficient (dimensionless, related to the housing system, manure storage or landspreading techniques) representing the proportion of TAN or Total nitrogen emitted to air. VC are derived from measurements designed and performed according to a national or an international protocol (e.g. VERA protocol) and validated for a farm with an identical type of technique and similar climatic conditions. Alternatively, information to derive VC can be taken from European or other internationally recognised guidance. The mass balance considers in particular any significant change to the type of livestock reared at the farm and/or to the techniques applied for housing, storage and landspreading. The monitoring technique is further developed at Member State level. In order to apply the monitoring technique, farms need to keep record of information on key parameters (according to the methodology defined by the Member State), e.g. numbers of animals, feed intake, crude protein content in the diet, quantity of manure generated, on-farm manure management practices.
b	Emission factors	Ammonia and dust emissions can be estimated on the basis of emission factors derived from measurements designed and performed according to a national or an international protocol (e.g. VERA protocol) on a farm with an identical type of technique (related to the housing system, manure storage and/or landspreading) and similar climatic conditions. Additionally, there are international guidance documents (e.g. the EMEP/EEA emission inventory guidebook) providing a tiered approach for the calculation of ammonia emissions at national level. This approach consists of three possible methods (Tier 1, Tier 2 and Tier 3) with increasing levels of complexity and accuracy. The Tier selection depends on the availability of data, significance of the livestock sector in the region, and the level of detail required. Tier 1 - This approach is a simple method that uses a single default emission factor (EF) to estimate ammonia emissions from particular animal categories. The EF is multiplied by the average annual population of each animal category to calculate emissions. The default Tier 1 EFs are calculated using Tier 2 default NH₃-N EFs, default N excretion data, and default data on proportions of total ammoniacal nitrogen (TAN) in excreta. Tier 2 - This approach uses mass flow to estimate ammonia emissions from livestock. It is based on the concept of a flow of total ammoniacal nitrogen (TAN) through the manure management system, which is composed of animal housing systems or holding areas, manure storage, and field application of manure. This approach requires more input data than Tier 1, including country-specific activity data, N excretion rates, and proportions of TAN in manure. Tier 2 estimates the amount of TAN available at each stage of manure management and calculates emissions using a mass balance approach. This approach is more accurate than Tier 1 and can account for differences in management systems and resulting emissions. It also allows for the inclusion of abatement measures and can be used to estimate emissions from multiple livestock categories. Tier 3 - This approach is an advanced method that uses emission modelling and installation data to estimate ammonia emissions from livestock. It is based on the use of country-specific emission factors (EFs) or the inclusion of abatement measures. Tier 3 can utilise the calculation procedure outlined under Tier 2, but with the use of country-specific EFs or the inclusion of abatement measures. This approach requires a high level of detail and accuracy, including data on facility-specific emissions, abatement measures, and management practices. It must be well documented and accompanied by supporting literature to ensure transparency and accuracy and can provide highly accurate estimates of

	Technique	Description
		ammonia emissions from livestock. For the purposes of ammonia emissions monitoring, a monitoring technique based on Tier 3 is to be developed at Member State level. In order to apply the monitoring technique, farms need to keep record of information on key parameters (according to the methodology defined by the Member State), e.g. numbers of animals, feed intake, crude protein content in the diet, quantity of manure generated, on-farm manure management practices.

4.8.3. Monitoring of methane emissions emissions to air

	Technique	Description
a	Emission factors	Methane emissions are calculated using emission factors, based on national or international methods (e.g. IPCC Guidelines for National Greenhouse Gas Inventories). These include the following approaches: Tier 1 – This approach entails multiplying the total amount of volatile solids (amount of manure excreted) from the animals by an emission factor specific to that type of animal category in a particular climate zone and using a specific manure management system. Tier 2 – This approach involves gathering detailed data on the animal categories and manure handling practices to create customised emission factors. In contrast to Tier 1, this approach allows for more precise calculations by using country-specific data rather than default values, and offers a detailed representation of manure management systems by livestock category and temperature regime (climate zone). Tier 3 – This approach involves the use of customised models or measurement-based methods to estimate methane emissions from manure management. The monitoring technique is further developed at Member State level. In order to apply the monitoring technique, farms need to keep record of information on key parameters (according to the methodology defined by the Member State), e.g. numbers of animals, feed intake, quantity of manure generated, on-farm manure management practices.

4.8.4. Monitoring of dust emissions to air

a	Emission factors	See technique (b) in Section 4.8.2.
---	------------------	-------------------------------------

4.8.5. Monitoring of the performance of air cleaning systems

	Technique	Description
a	Verification of the air cleaning system performance by measuring ammonia, odour and/or dust under practical farm conditions and according to a prescribed measurement protocol and using EN standard methods or other national or international methods ensuring data of an	The verification is done by measurement of ammonia, odour and/or dust under practical farm conditions and according to a prescribed measurement protocol and using EN standard methods or other national or international methods ensuring data of an equivalent scientific quality. Normally, the measurements are done at the inlet and outlet air and for all additional parameters relevant for operation (e.g. air flow rate, pressure drop, temperature, pH level, conductivity). Measurements are performed under summer climatic conditions (a period of at least eight weeks with a ventilation rate > 80 % of the maximum design ventilation rate) and winter climatic conditions (a period of at least eight weeks with a ventilation rate < 30 % of the maximum design ventilation rate), with representative

	Technique	Description
	equivalent scientific quality.	management and full capacity of the housing and only if an adequate time period (e.g. four weeks) has elapsed after the last change of wash water. Different sampling strategies can be applied.
b	Control of the effective function of the air cleaning system (e.g. by continuously recording operational parameters or using alarm systems).	Operation of an electronic logbook in order to record all measuring and operational data over a period of at least 5 years. Recorded parameters depend on the type of air cleaning system and may include: (a) pH and conductivity of the scrubbing liquid; (b) air flow and pressure drop of the abatement system; (c) pump operating time; (d) water and acid consumption. Other parameters can be recorded manually. The registered data are typically made available to the competent authority (e.g. electronically).

4.8.6. Monitoring of process parameters

	Technique	Description
a	Monitoring of water consumption at farm level and for the main water-consuming processes.	Recording using meters or invoices for example. The main water-consuming processes (e.g. cleaning, feeding) are monitored separately.
b	Monitoring of electric energy consumption.	Recording using meters or invoices for example. The electricity consumption of animal houses is monitored separately. The main energy-consuming processes (heating, ventilation, lighting, etc.) can also be monitored separately.
c	Monitoring of fuel consumption.	Recording using meters or invoices for example.
d	Monitoring of the number of animals including births and deaths.	Recording of incoming and outgoing animals, including births and deaths, using registers for example.
e	Monitoring of feed consumption.	Recording using invoices or registers for example.
f	Monitoring of the quantity of manure generated.	Recording using registers for example.

5. EMERGING TECHNIQUES

5.1. Ammonia and methane emissions from animal houses

	Technique	Description	Information on emissions
a	Adsorption of ammonia from gaseous media	Acidic, porous adsorbents (e.g. clays modified with metal oxides) are used to remove ammonia from the air and slurry. For air treatment, adsorbents are installed in ventilation columns, where ammonia, from the air which passes through the column, binds to the adsorbent via chemisorption or physisorption. Some adsorbents allow regeneration via thermal or vacuum desorption, enabling their reuse, although their cost may be higher and farm energy consumption may be increased. For slurry treatment, adsorbents can be directly added to pits or storage facilities, without the need for any specific modification to the housing structure (e.g. installation of an air ventilation system with columns) and without additional energy consumption for their reuse. However, since these non-regenerable materials stay in the solid slurry phase, they must not contain any toxic substances.	An ammonia reduction of at least 38 % is expected to be achieved in comparison to a reference system without the use of this technique.
b	Alternative litter materials	Alternative litter materials are used, with favourable characteristics in terms of prevention of ammonia generation and the consequent decrease in emissions in comparison to conventional litter materials (e.g. straw, pellets, wood shavings). The emission-reducing properties can be enhanced by the use of additives in conventional materials. By changing the chemical or microbial characteristics of the litter, the processes responsible for ammonia generation can be slowed down. The choice of litter material and its additives needs to take into account their subsequent use or disposal.	N.A.
c	Use of other slurry additives	In addition to the slurry additives described in Section 4.2.1, the use of other slurry additives (e.g. urease inhibitor PPDA, aqueous solution of H ₂ O ₂) has been identified as an emerging technique.	For addition of urease inhibitor PPDA, ammonia reductions of at least 28 % (if applied under the slatted floor) or 32 % (if applied on the solid floor) are expected to be achieved in comparison to a reference system of deep pit with fully slatted floor without any additional techniques. For addition of 35 to 50 % aqueous solution of H ₂ O ₂ , an ammonia reduction of at least 50 % is expected to be achieved in comparison to a reference system of deep pit with fully slatted floor without any additional techniques.

	Technique	Description	Information on emissions
d	Robot for removing slurry	Slurry vacuum robots are autonomous devices operating under the slatted floors in pig houses. They aim to remove slurry continuously, thus decreasing ammonia emissions and odour, as well as providing better comfort for the pigs and farm employees. The frequency of robots' passages can be adjusted (e.g. several times a day). In order to implement the technique in existing farms, a minimum clearance under the grating is needed (e.g. more than 80 cm), the floor slope needs to be limited (e.g. less than 10 %), and drilling through walls between rooms must be possible.	An ammonia reduction of at least 28 % is expected to be achieved in comparison to a reference system of deep pit with fully slatted floor without any additional techniques. Methane emissions are also expected to be reduced.

5.2. Manure storage

	Technique	Description
a	Flaring of methane from slurry storage	Methane-rich gases collected from covered slurry stores are diverted to a low-pressure flare that oxidises methane to carbon dioxide and water vapour, thereby significantly reducing the climate impact compared to direct atmospheric release of methane. The technique requires slurry tanks to be equipped with systems such as dense tent covers, in order to avoid air in-leaks and maintain the methane concentration above 5 %. The gases are extracted using a vacuum pump, then pass through a condensate trap (the captured condensates are pumped back to the slurry storage) before reaching the burner, while the combustion air is added just below the burner head. Instruments such as spark igniters, as well as flame, temperature and excess oxygen sensors, are used to adjust the process to optimal combustion conditions. The energy in the combustion process can be partially recovered using heat exchangers, if a demand exists on farm. This technique does not require the use of additional fuel for pilot flame.

5.3. On-farm processing of manure

	Technique	Description
a	Phosphorus recovery from manure	Phosphate salts are precipitated through the addition of ions (e.g. magnesium and potassium) and a pH regulator, often in a reactor. To optimise the process and minimise unwanted co-precipitation, it is necessary to monitor the solution's pH and the molar ratios of the ions present in the solution during the process. Phosphorus can be recovered in the form of salts (e.g. struvite) which can be reused as fertilisers.
b	Resource recovery from digestates by using gypsum	Digestate produced on farm is treated to obtain mineral fertilisers by using gypsum, which can be "natural" or "synthetic", i.e. generated in flue-gas desulphurisation systems (gas cleaning systems used for capturing sulphur dioxide emissions from flue-gases). When gypsum is added to digestate, ammonia and carbon dioxide are gradually stripped at low temperatures/pressures (e.g. 50 to 85°C / 200 to 700 mbar, without air or steam). Precise adjustment of the chemical and thermodynamic equilibria in each step ensures low power and heat consumption and a highly efficient process. In order to decrease the power consumption, waste heat from another process can be used. The formed suspension is separated using a filter press into two mineral fertilisers: ammonium sulphate (solution) and calcium carbonate (solid).

	Technique	Description
c	Gas-permeable membranes (GPM) for ammonia capture from liquid media	This technique operates in two steps: in the first step, gaseous ammonia is released from ammonia-rich liquid streams (e.g. anaerobic digestate, slurry) by aeration and spraying, generating an ammonia-rich air stream; in the second step, this gas stream comes into contact with a microporous, hydrophobic membrane, where ammonia passes through the pores of the membrane. The driving force for ammonia transfer across membranes is the concentration gradient. The process is carried out at atmospheric pressure. Once the ammonia is in the acidic solution (on the other side of the membrane), it combines with free protons to form non-volatile ammonium ions (NH_4^+). Various inorganic or organic acids can be used to prepare the acidic solution, such as sulphuric, hydrochloric, phosphoric and lactic acid. In this way, ammonia contained in the liquid is transformed into a valuable ammonium fertiliser (e.g. ammonium sulphate, ammonium chloride, ammonium phosphate, ammonium lactate). The system is usually located outside the housing and needs sufficient space for maintenance. An ammonia reduction of at least 30 % is expected to be achieved in comparison to a reference system without the use of this technique.

5.4. Monitoring

	Technique	Description
a	Calculation based on real time measurement of ammonia and/or methane concentration in the air and of the ventilation flow.	Using sensors, the concentration of ammonia and methane in the outgoing ventilation air from animal houses is measured continuously. The ventilation flow rate is measured using measuring fans or frequency-controlled operating fans with recording. Both measurements are performed according to a strict protocol, based on scientifically established methods. By multiplying the measured concentration by the ventilation flow rate, the emission from the animal housing is calculated. This monitoring technique can be used in plants with forced ventilation.